

Introduction to the strongly-correlated electron problem

Lecture 1- Introduction to the Hubbard model

Why

Introduction to the strongly-correlated electron problem



Are these things
even related?

Summer school on fault-tolerant quantum computing

Why

Heard very often: fault-tolerant quantum computers will help us...

Design new materials

Applications:

Solar cells

Batteries

Magnetic memories

Catalysts

Understand the
properties of quantum
matter

Areas:

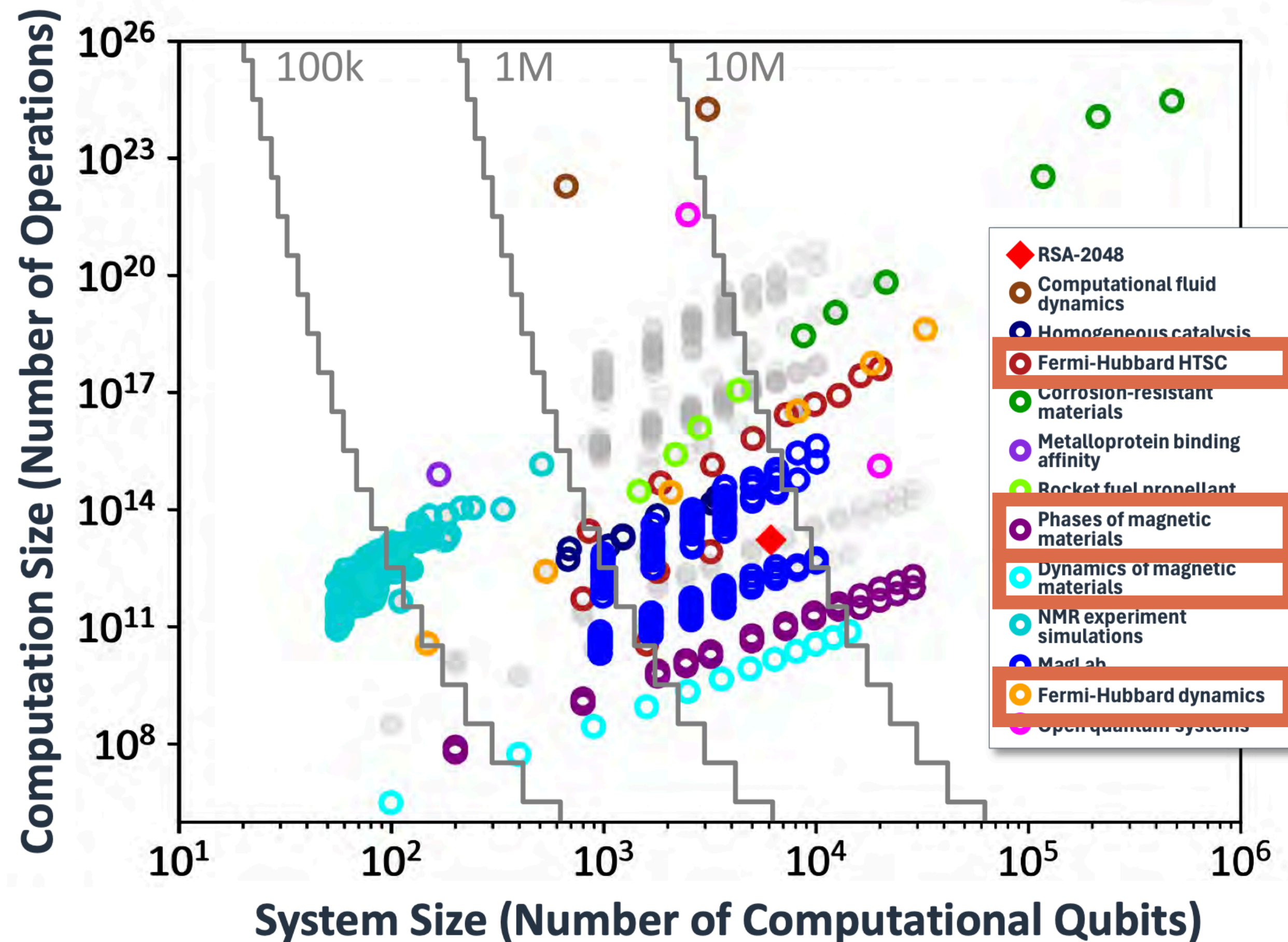
Quantum chemistry

High-temperature superconductivity

Out of equilibrium physics

(not exhaustive...)

Why



Grey solid circles represent pessimistic resource estimates. Colored circles are optimistic resource estimates based on known improvements. All points supported by detailed published pre-prints.

Altepeter (DARPA, 2024)

Specific example: Quantum Benchmarking Initiative from DARPA

What about a more fundamental point of view?

Outline

- Lecture 1: introduction to the strongly-correlated electron problem
 - What it is, why should we care
 - What we know about the Hubbard model
- Lecture 2: Quantum computing for the Hubbard model
 - From fermions to qubits
 - Using a quantum processor to "solve" the Hubbard model

Outline

Lecture 1

- What is the many-body problem in quantum materials
- Why is this problem still relevant today
- The Hubbard model
- Known properties of the model in 2D
- Open questions

Goal of the lecture:

Understand **what the Hubbard model is** and **what we currently know** about it *before* trying to solve it with a quantum processor

What is the many-body problem
in quantum materials

A very general problem

Generic hamiltonian of N electrons in a lattice: $H = H_{\text{kin}} + V$

$$H_{\text{kin}} = - \sum_{i=1}^{N_e} \frac{\hbar^2}{2m} \nabla_{\mathbf{r}_i}^2 - \sum_{l=1}^{N_{\text{ion}}} \frac{\hbar^2}{2M} \nabla_{\mathbf{R}_l}^2$$

Kinetic part: one-body terms

$$V = \sum_{i < j} \frac{e^2}{4\pi\epsilon_0 |\mathbf{r}_i - \mathbf{r}_j|} + \sum_{k < l} \frac{Z_k Z_l e^2}{4\pi\epsilon_0 |\mathbf{r}_i - \mathbf{r}_j|} - \sum_{l=1}^{N_{\text{ion}}} \sum_{i=1}^{N_{\text{el}}} \frac{Z_l e^2}{4\pi\epsilon_0 |\mathbf{R}_l - \mathbf{r}_i|}$$

Potential part: two-body terms

A very general problem - simplified

Generic hamiltonian of N electrons in a lattice: $H = H_{\text{kin}} + V$

$$H_{\text{kin}} = - \sum_{i=1}^{N_e} \frac{\hbar^2}{2m} \nabla_{\mathbf{r}_i}^2$$

Kinetic part: one-body terms

$$V = \sum_{i < j} \frac{e^2}{4\pi\epsilon_0 |\mathbf{r}_i - \mathbf{r}_j|} - \sum_{l=1}^{N_{\text{ion}}} \sum_{i=1}^{N_{\text{el}}} \frac{Z_l e^2}{4\pi\epsilon_0 |\mathbf{R}_l - \mathbf{r}_i|}$$

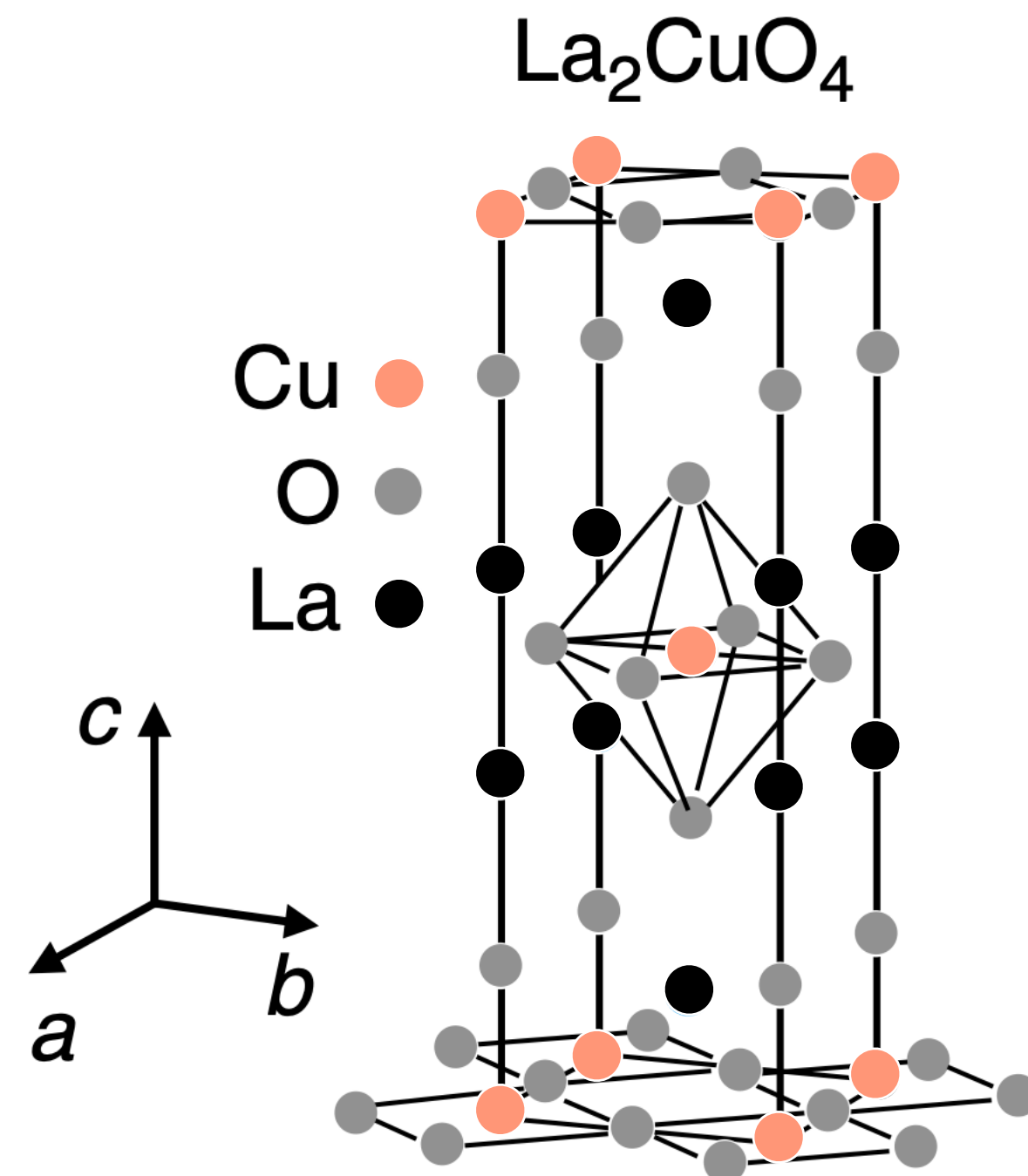
Potential part: two-body terms

Impossible to solve exactly

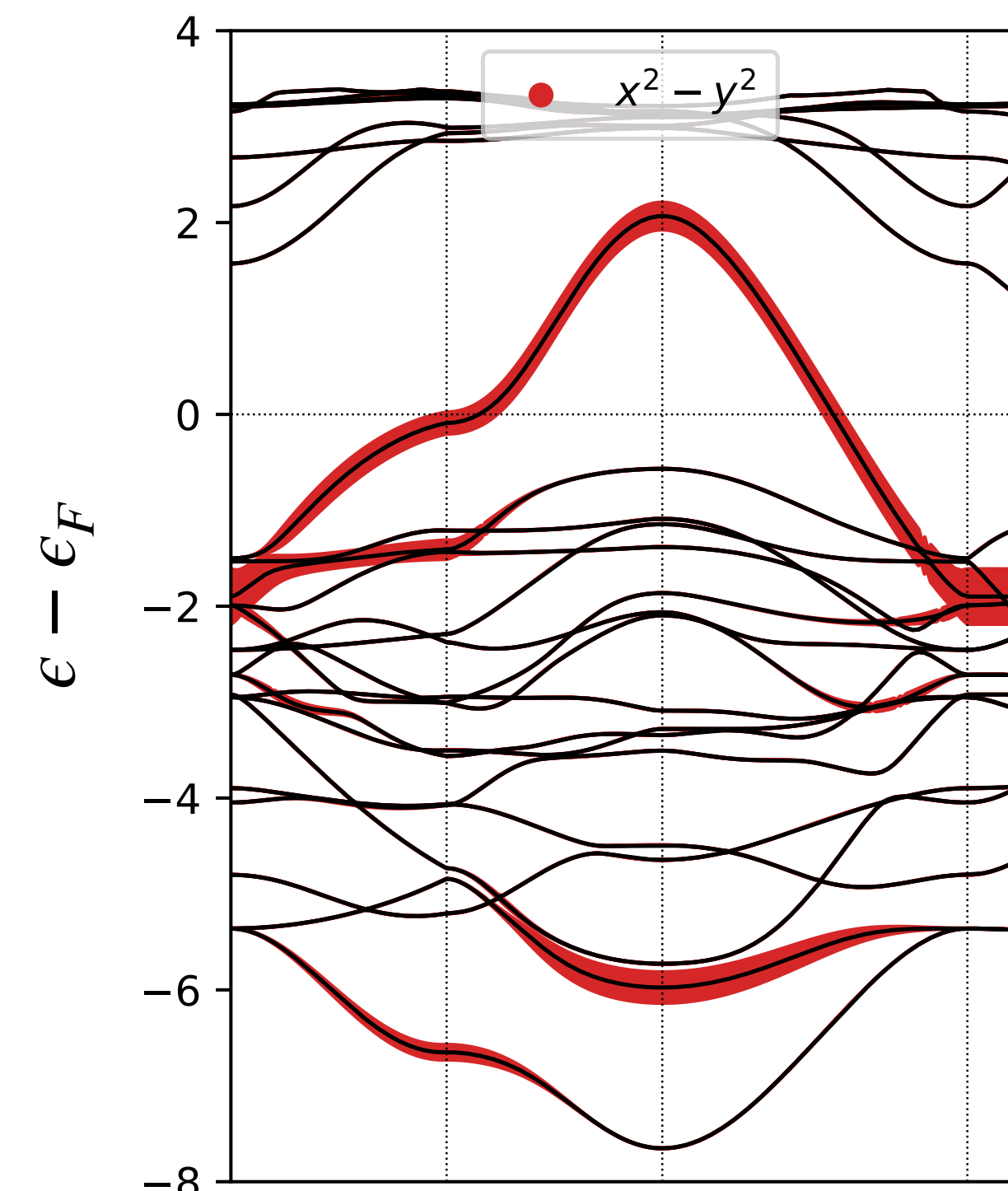
The traditional route: density functional theory

Principle of DFT: calculate the ground state properties of a system from **first principles**

Input: crystal structure



Output: ground state electronic properties

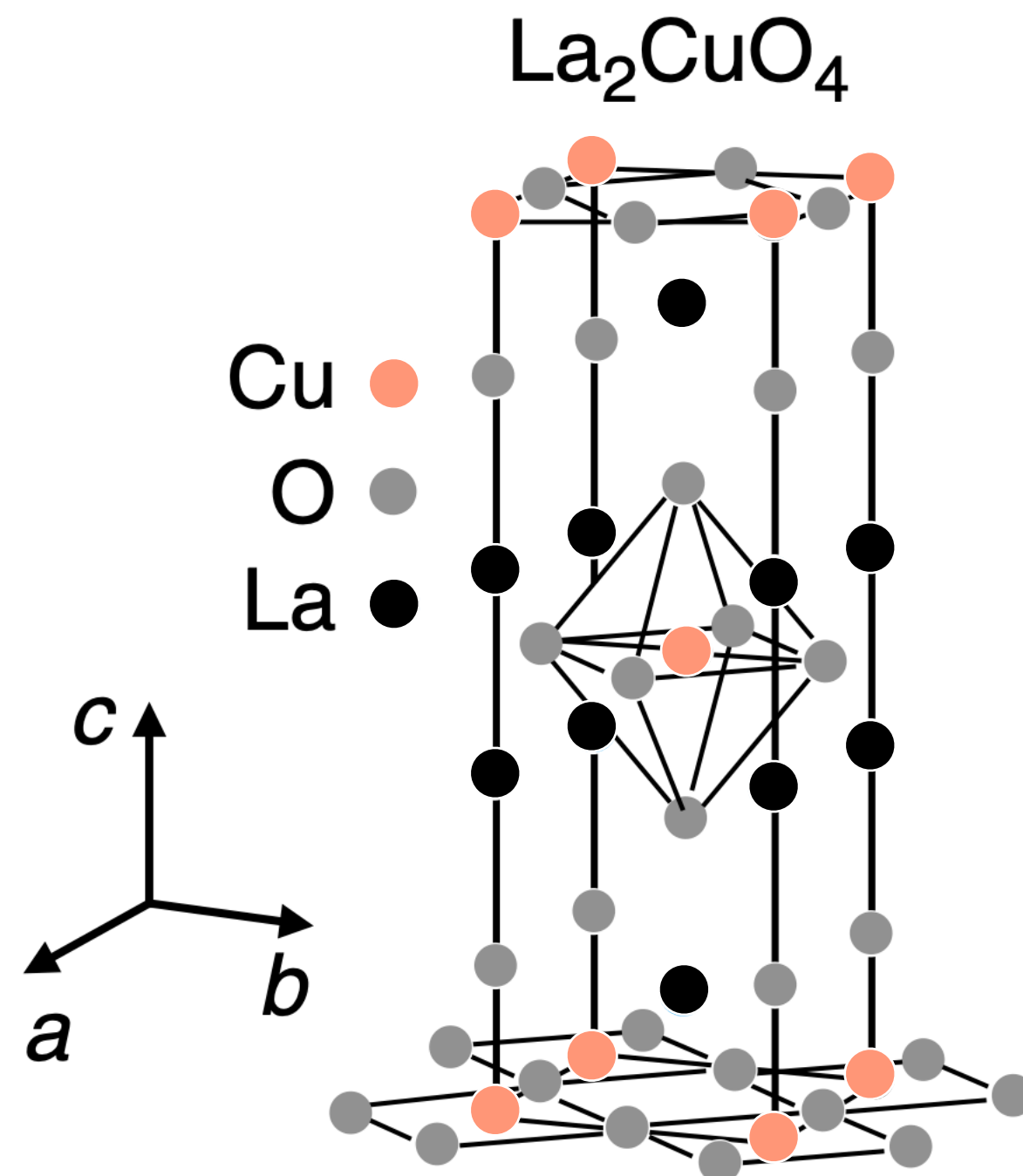


Ground state energy
Magnetic properties
Electronic gap
Phonon dispersion
...

The traditional route: density functional theory

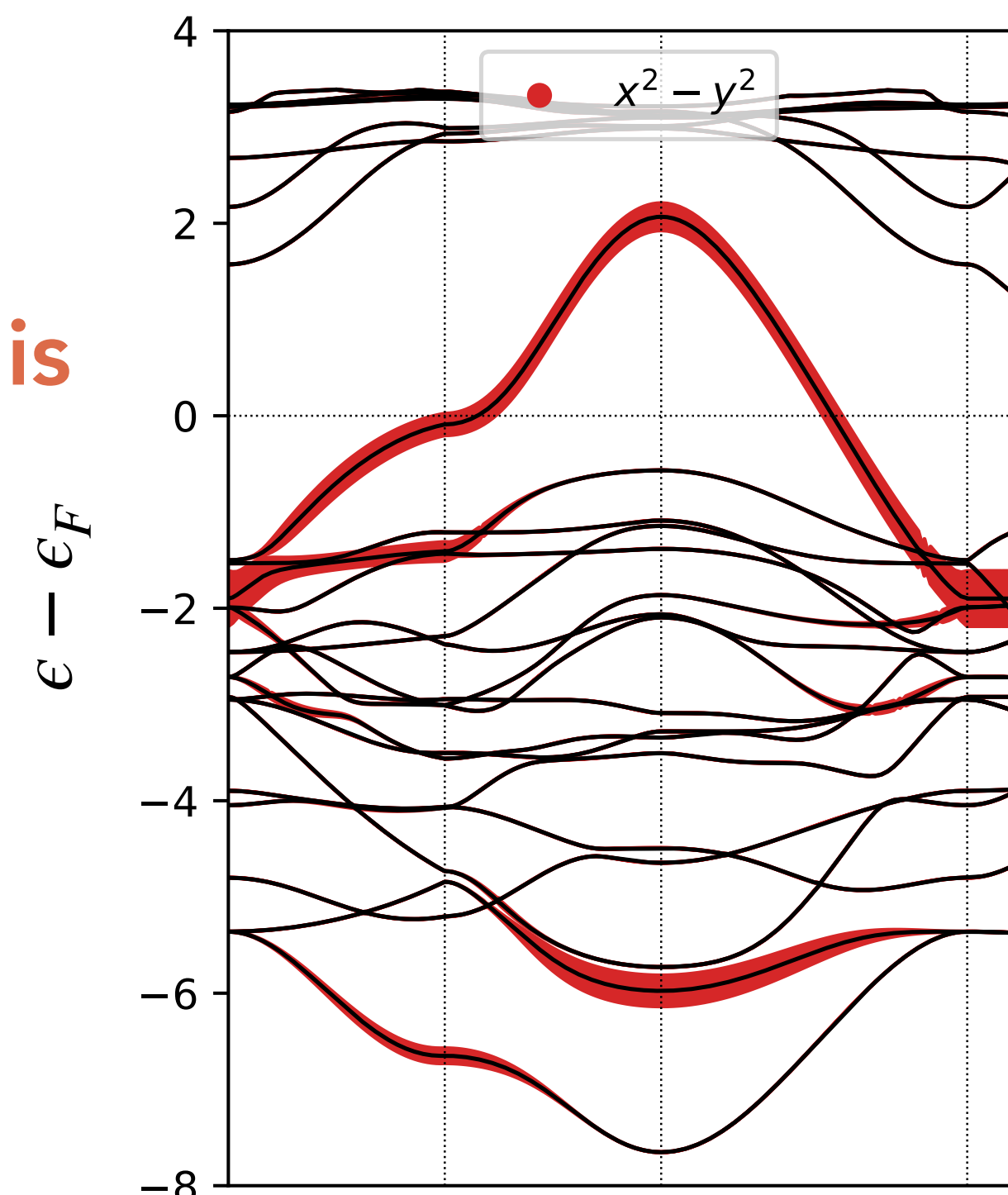
Principle of DFT: calculate the ground state properties of a system from **first principles**
Treat the electron-electron interaction approximately

Input: crystal structure



Output: ground state electronic properties

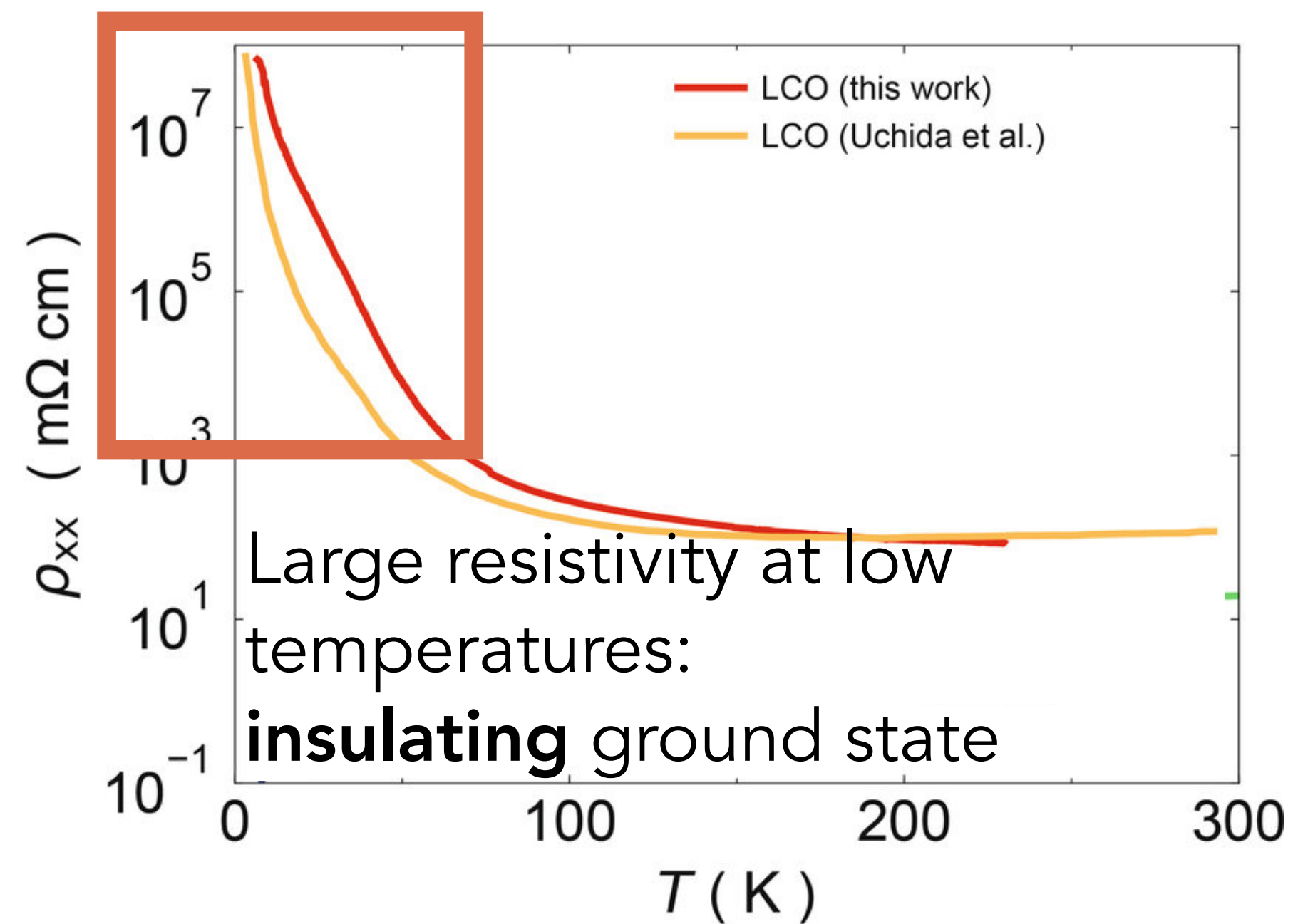
**But sometimes, DFT is
(very) wrong**



Ground state energy
Magnetic properties
Electronic gap
Phonon dispersion
...

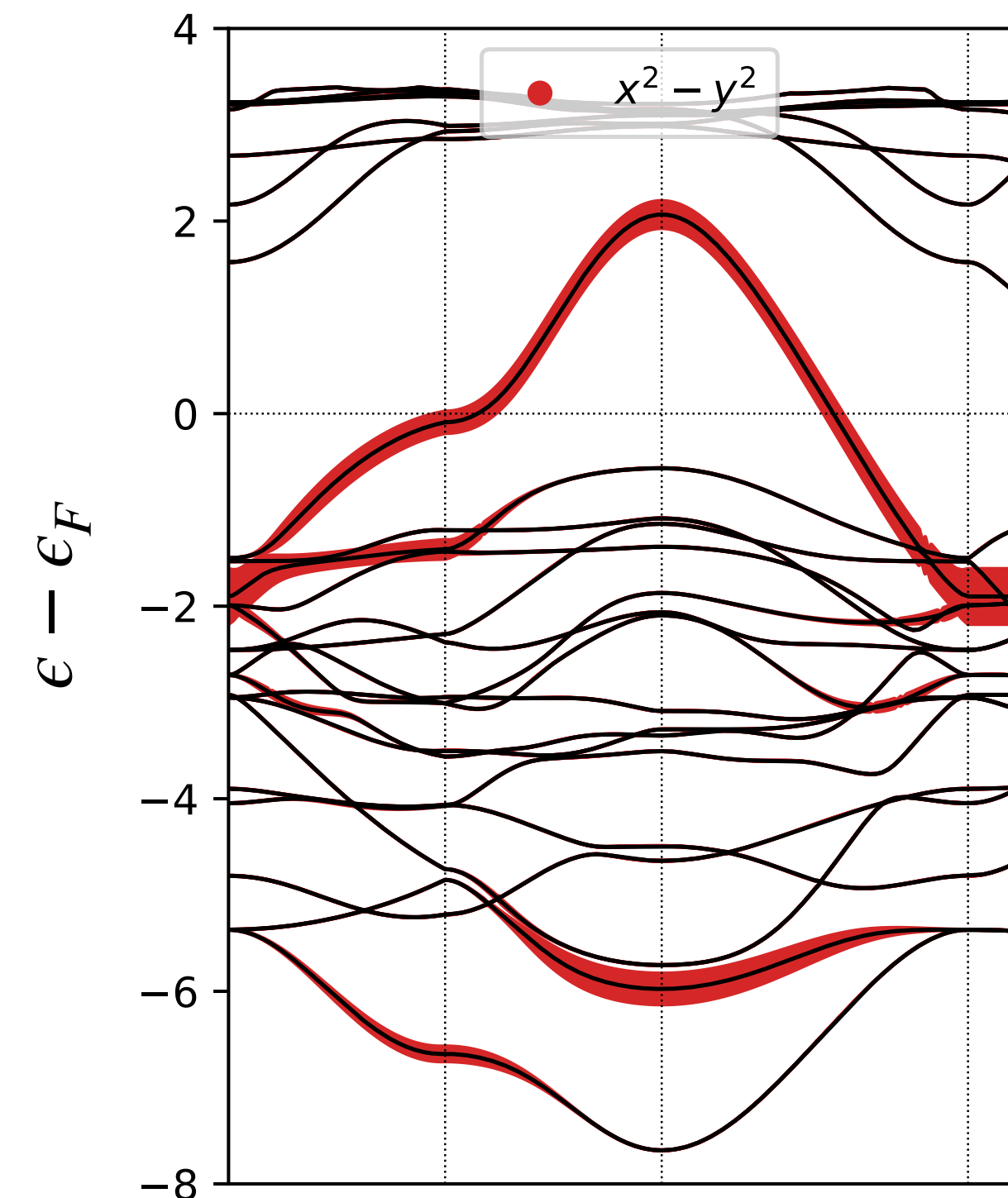
When DFT doesn't work: an example

Resistivity measurements



G. Grissonnanche *et al.*
Nature 571, 376–380 (2019)

DFT output: ground state electronic properties



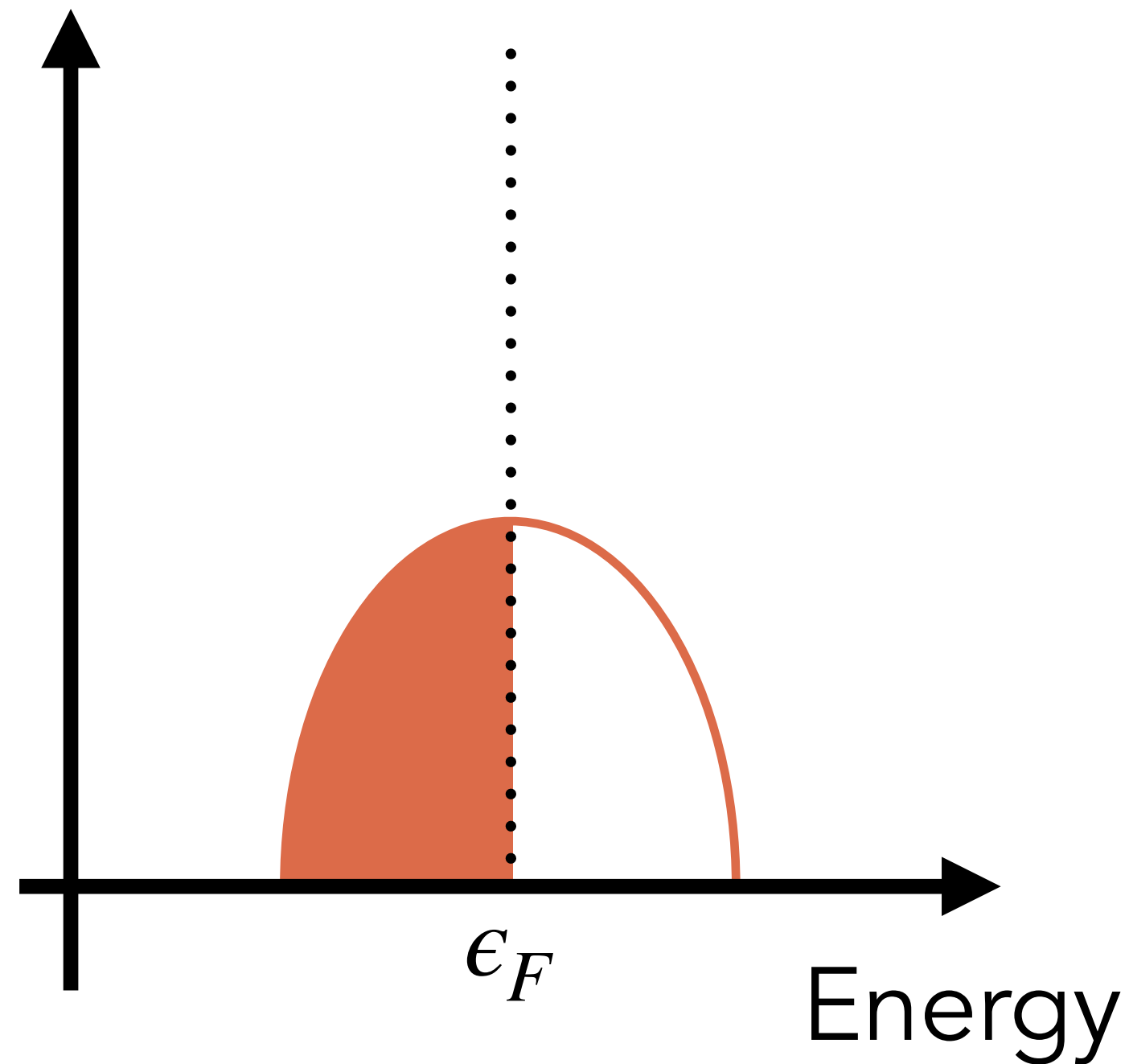
Metallic ground state

Interactions lead to new phenomena

Key phenomenon in strongly-correlated electron systems

DFT prediction

Density of states



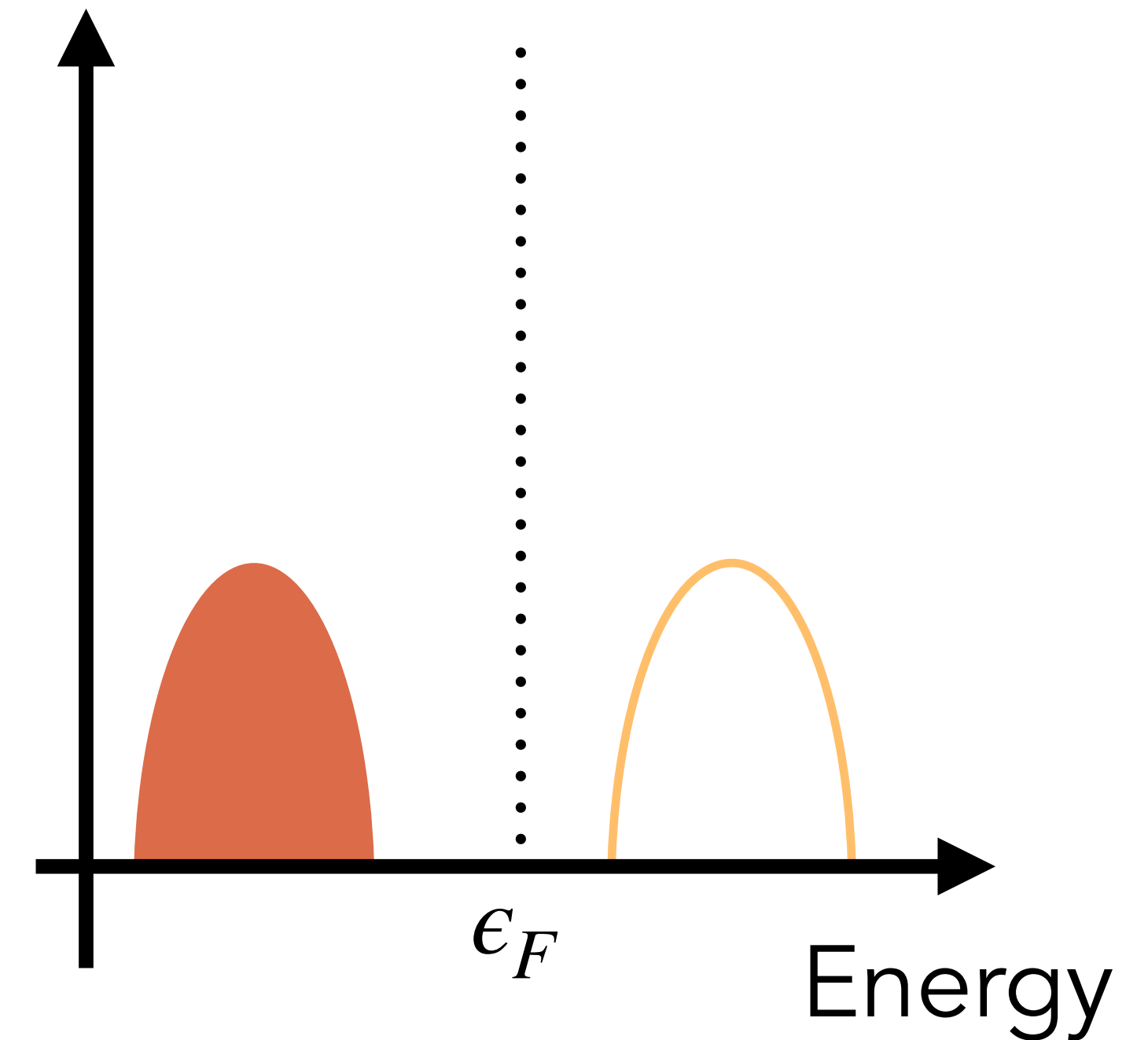
Metal

Interactions
between
electrons



Experimental observation

Density of states



Mott Insulator

A definition of strong correlations

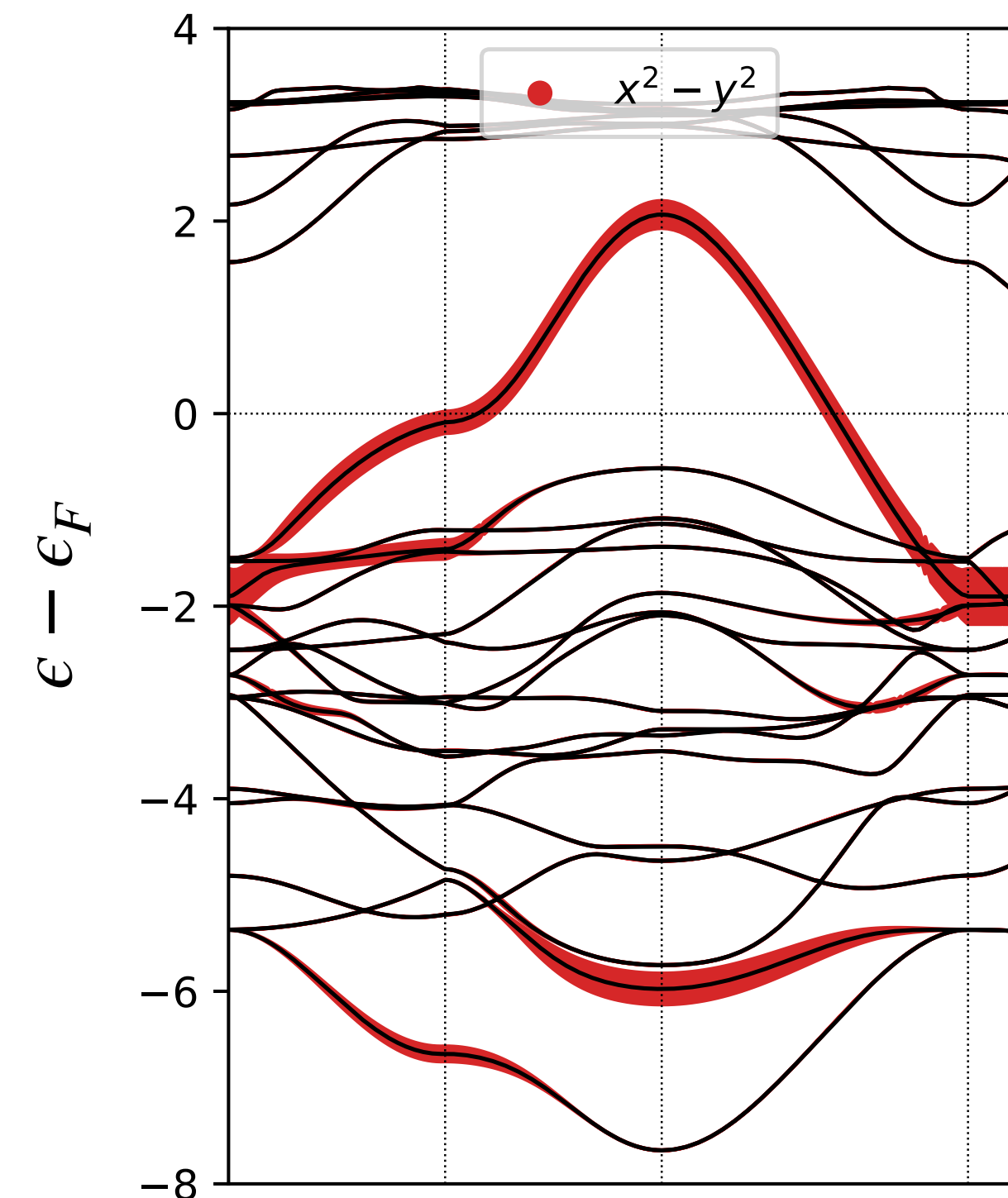
What is missing

Missing here: a proper treatment of electron-electron interactions

Definition: strong correlations

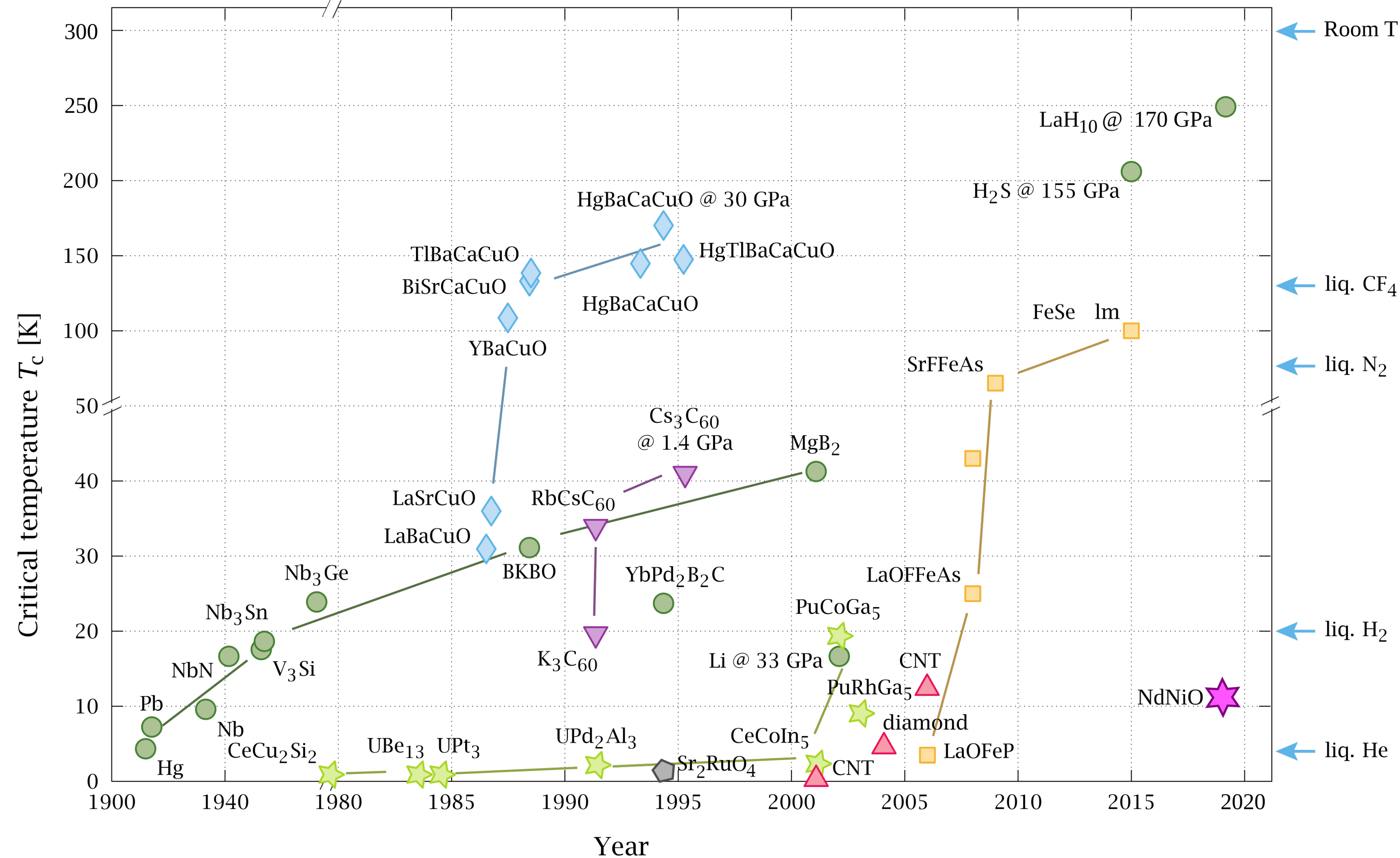
Strong electronic correlations: effects from electron-electron interactions that are not treated at the DFT level.

DFT output: ground state electronic properties



Metallic ground state

Examples: strongly correlated materials



**High-temperature
superconductors**

Heavy fermions

**Correlated
topological
materials**

**Some magnetic
materials**

...

2. The Hubbard model

A model to study **strong electronic correlations**

Reminders: states with multiple fermions

First quantization

ONE FERMION

Single-particle basis states: $|\alpha_i\rangle$

Some combination of quantum numbers: spin, orbital index...

Space: V_1

Single-particle wave function: $\phi_{\alpha_1}(\mathbf{r}) = \langle \mathbf{r} | \alpha_1 \rangle$

TWO FERMIONS

Tensor product state: $|\alpha_1\rangle \otimes |\alpha_2\rangle$

Antisymmetrized basis states: $|\alpha_1, \alpha_2\rangle = \frac{1}{\sqrt{2}} (|\alpha_1\rangle \otimes |\alpha_2\rangle - |\alpha_2\rangle \otimes |\alpha_1\rangle)$

Pauli principle

Space: $V_2 = \mathcal{A}(V_1 \otimes V_1)$

Only the antisymmetric states

Reminders: states with multiple fermions

First quantization

N FERMIONS

With N fermions:

$$|\alpha_1, \dots, \alpha_N\rangle = \frac{1}{N!} \sum_{p \in S_N} \epsilon_p |\alpha_{p(1)}\rangle \otimes \dots \otimes |\alpha_{p(N)}\rangle$$

permutations $\xrightarrow{\quad}$ $p \in S_N$ $\xrightarrow{\quad}$ sign of the permutation

Space: $V_N = \mathcal{A}(V_1 \otimes V_1 \otimes \dots \otimes V_1)$

$\xrightarrow{\quad}$ only the antisymmetric states

Ok. But in practice, what if the number of particles is very large or not fixed (or both)?

Second quantization

Fock space

Second quantization

FOCK SPACE

Direct sum of the spaces that contain a fixed number of particles

$$V = V_0 \oplus V_1 \oplus V_2 \oplus \dots$$

States in V can contain any number of indistinguishable particles

$$|\alpha_1, \dots, \alpha_N\rangle = \frac{1}{N!} \sum_{p \in S_N} \epsilon_p |\alpha_{p(1)}\rangle \otimes \dots \otimes |\alpha_{p(N)}\rangle$$

Second quantization

Creation and annihilation operators

Second quantization

CREATION OPERATOR

Introduce a creation operator: $c_{\alpha_i}^\dagger$

Effect: $c_{\alpha_i}^\dagger |0\rangle = |\alpha_i\rangle$ creates a particle in state α_i

$c_{\alpha_i}^\dagger |\alpha_i\rangle = 0$ fermions: at most one particle in one state

ANNIHILATION OPERATOR

Introduce an annihilation operator: c_{α_i}

Effect: $c_{\alpha_i} |\alpha_i\rangle = |0\rangle$ destroys a particle in state α_i

$c_{\alpha_i} |0\rangle = 0$ destroys the vacuum

Second quantization

Creation and annihilation operators

Second quantization

CREATION OPERATOR

Introduce a creation operator: $c_{\alpha_i}^\dagger$

Effect: $c_{\alpha_i}^\dagger |\alpha_1, \dots, \alpha_N\rangle = |\alpha_i, \alpha_1, \dots, \alpha_N\rangle$ important: still antisymmetric

ANNIHILATION OPERATOR

Introduce an annihilation operator: c_{α_i}

Effect: $\langle \alpha_1, \dots, \alpha_N | c_{\alpha_i} = \langle \alpha_i, \alpha_1, \dots, \alpha_N |$

important: still antisymmetric

Requires **anticommutation**
relations:

$$\{c_\alpha^\dagger, c_\beta\} = c_\alpha^\dagger c_\beta + c_\beta c_\alpha^\dagger = \delta_{\alpha\beta}$$

$$\{c_\alpha^\dagger, c_\beta^\dagger\} = 0 \quad \{c_\alpha, c_\beta\} = 0$$

Second quantization

Creation and annihilation operators

Second quantization

DEFINITION OF THE STATES

$$|\alpha_1, \dots, \alpha_N\rangle = c_{\alpha_1}^\dagger \dots c_{\alpha_N}^\dagger |0\rangle$$

vacuum



NUMBER OPERATOR

$$\hat{n}_{\alpha_i} = c_{\alpha_i}^\dagger c_{\alpha_i}$$
$$\hat{n}_{\alpha_i} |\alpha_1, \dots, \alpha_N\rangle = \begin{cases} 0 & \text{if } \alpha_i \text{ is not in the state,} \\ |\alpha_1, \dots, \alpha_N\rangle & \text{if } \alpha_i \text{ is in the state} \end{cases}$$

eigenvalues: 0 and 1



Fermions: each state $|\alpha_i\rangle$ is either **empty** or **occupied once**

Second quantization

Operators

Second quantization

ONE-BODY OPERATORS

one-body operator defined in V_1

$$\hat{O} = \sum_{\alpha_i, \alpha_j} \langle \alpha_i | \hat{O} | \alpha_j \rangle | \alpha_i \rangle \langle \alpha_j | = \sum_{\alpha_i, \alpha_j} O_{ij} c_{\alpha_i}^\dagger c_{\alpha_j}$$

using closure relations $\xrightarrow{\quad}$ $O_{ij} = \langle \alpha_i | \hat{O} | \alpha_j \rangle$

$$\begin{aligned} | \alpha_i \rangle \langle \alpha_j | \alpha_m \rangle &= \delta_{jm} | \alpha_i \rangle \\ &= \delta_{jm} c_{\alpha_i}^\dagger | 0 \rangle \end{aligned}$$

use $| \alpha_i \rangle = c_{\alpha_i}^\dagger | 0 \rangle$

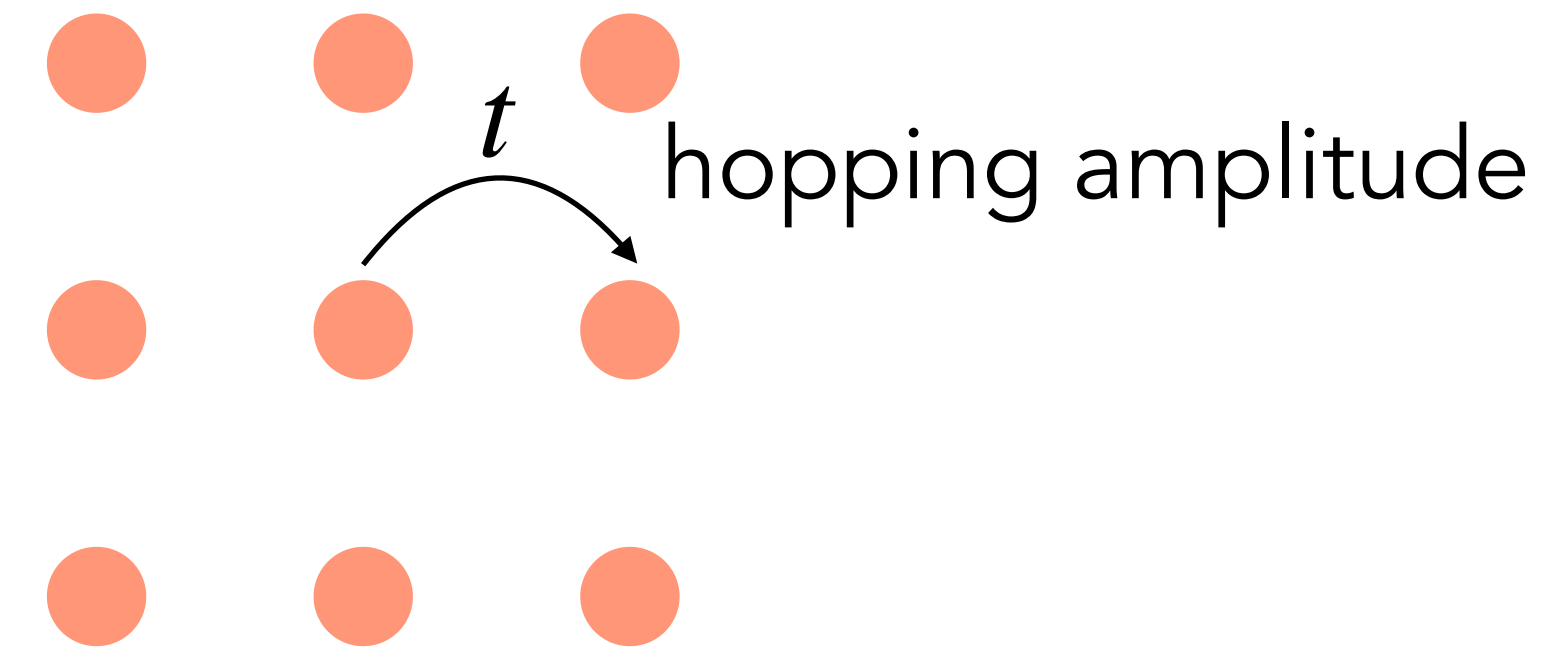
$$c_{\alpha_i}^\dagger c_{\alpha_j} = | \alpha_i \rangle \langle \alpha_j | \quad \text{in } V_1$$

also: $c_{\alpha_i}^\dagger c_{\alpha_j} | \alpha_m \rangle = \delta_{jm} c_{\alpha_i}^\dagger | 0 \rangle$

A simple tight-binding model in second quantization

$$H_0 = -t \sum_{\langle ij \rangle, \sigma} c_{i\sigma}^\dagger c_{j\sigma}$$

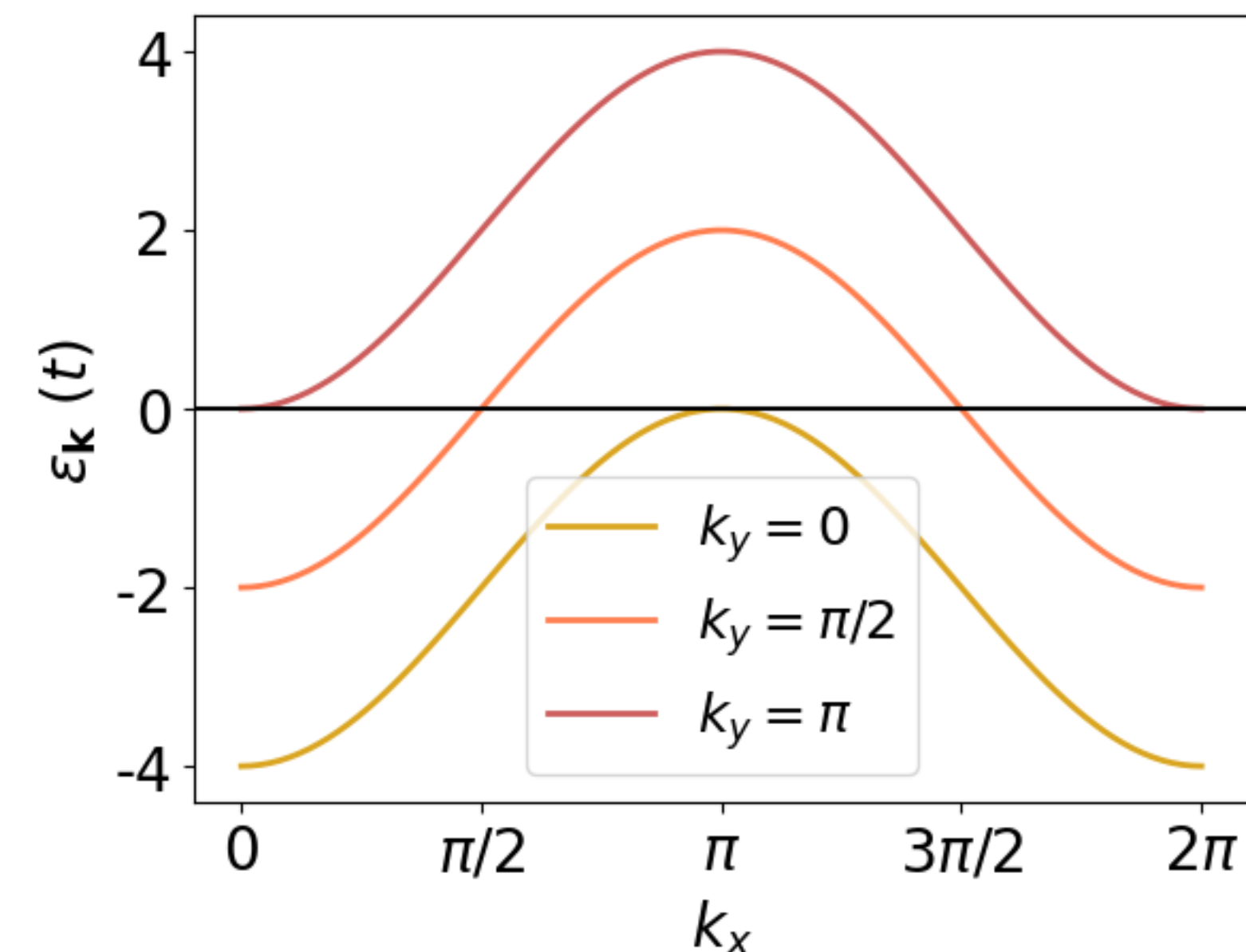
$\xrightarrow{\text{annihilate an electron of spin } \sigma \text{ on site } j}$
 $\xleftarrow{\text{create an electron of spin } \sigma \text{ on site } i}$



diagonalization:
Fourier transform

$$c_{\mathbf{k}\sigma} = \frac{1}{\sqrt{N}} \sum_i e^{-i\mathbf{k} \cdot \mathbf{r}_i} c_{i\sigma}$$

$$H_0 = \sum_{\mathbf{k}, \sigma} \epsilon(\mathbf{k}) c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma}$$



Second quantization

Operators

Second quantization

ONE-BODY OPERATORS

one-body operator defined in V_1

$$\hat{O} = \sum_{\alpha_i, \alpha_j} \langle i | \hat{O} | j \rangle c_{\alpha_i}^\dagger c_{\alpha_j}$$

TWO-BODY OPERATORS

two-body operator defined in V_2

$$\hat{V} = \sum_{i,j,k,l} \langle i, j | \hat{V} | k, l \rangle c_i^\dagger c_j^\dagger c_l c_k$$

The Hubbard model

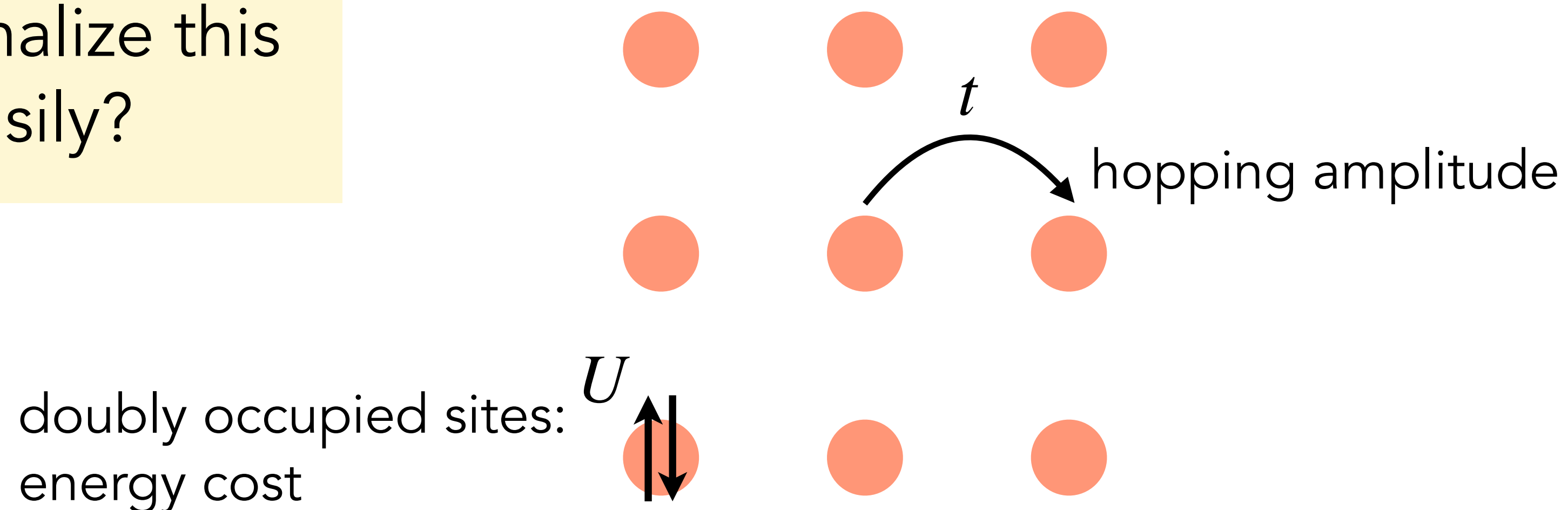
$$H = -t \sum_{\langle ij \rangle, \sigma} c_{i\sigma}^\dagger c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}$$

one-body operator

two-body operator

**Simplest toy-model to
study local electron-
electron interactions**

Can we diagonalize this
hamiltonian easily?



The Hubbard model

$$H = -t \sum_{\langle ij \rangle, \sigma} c_{i\sigma}^\dagger c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}$$

one-body operator

two-body operator

Can we diagonalize this hamiltonian easily?

$$H = \sum_{\mathbf{k}, \sigma} \epsilon(\mathbf{k}) c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}$$

The two terms are diagonal in **two reciprocal bases**

The Hubbard model

$$H = -t \sum_{\langle ij \rangle, \sigma} c_{i\sigma}^\dagger c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}$$

Hilbert space size

ONE SITE

$|0\rangle \quad |\uparrow\downarrow\rangle$

$|\uparrow\rangle \quad |\downarrow\rangle$

4 states

TWO SITES

$|0,0\rangle \quad |\uparrow\downarrow,0\rangle$

$|\uparrow,0\rangle \quad |\downarrow,0\rangle$

$|0,\uparrow\rangle \quad |\uparrow\downarrow,\uparrow\rangle$

... 16 states

...

N SITES

4^N states

The Hubbard model

$$H = -t \sum_{\langle ij \rangle, \sigma} c_{i\sigma}^\dagger c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}$$

Hilbert space size

	N	Matrix size
N SITES	4	256 x 256
4 ^N states	30	~10 ³⁶
	200	~10 ²⁴⁰
	10 ²³	...let's not think about this

Full diagonalization: limited to **small** systems, far from the thermodynamic limit

Some tricks:

- symmetries
- restricted spin sectors
- diagonalization methods (example: Lanczos)

The Hubbard model

Numerical
exercise

$$H = -t \sum_{\langle ij \rangle, \sigma} c_{i\sigma}^\dagger c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}$$

2 site Hubbard model

Write the hamiltonian in the following basis:

$$\hookrightarrow (|\uparrow\downarrow, 0\rangle, |\uparrow, \downarrow\rangle, |\downarrow, \uparrow\rangle, |0, \uparrow\downarrow\rangle)$$

Diagonalize the hamiltonian numerically for values of U between 0 and 10 (take t=1)

$$\hookrightarrow H|\psi_{\text{GS}}\rangle = E_{\text{GS}}|\psi_{\text{GS}}\rangle$$

Calculate and plot the double occupancy as a function of U

$$\hookrightarrow \langle D \rangle = \langle \psi_{\text{GS}} | \sum_i n_{i\uparrow} n_{i\downarrow} | \psi_{\text{GS}} \rangle$$

The Hubbard model

$$H = -t \sum_{\langle ij \rangle, \sigma} c_{i\sigma}^\dagger c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}$$

Does this mean that there are no exact solutions to the Hubbard model and that only a powerful fault-tolerant quantum computer could solve it fully?

...no, as we will see in the next section.

3. What we know about the 2D Hubbard model

Reminder: goal of the lecture

In order to know how a quantum computer could help "solving" the Hubbard model, one needs to know what is currently known about the model

Goal of the lecture:

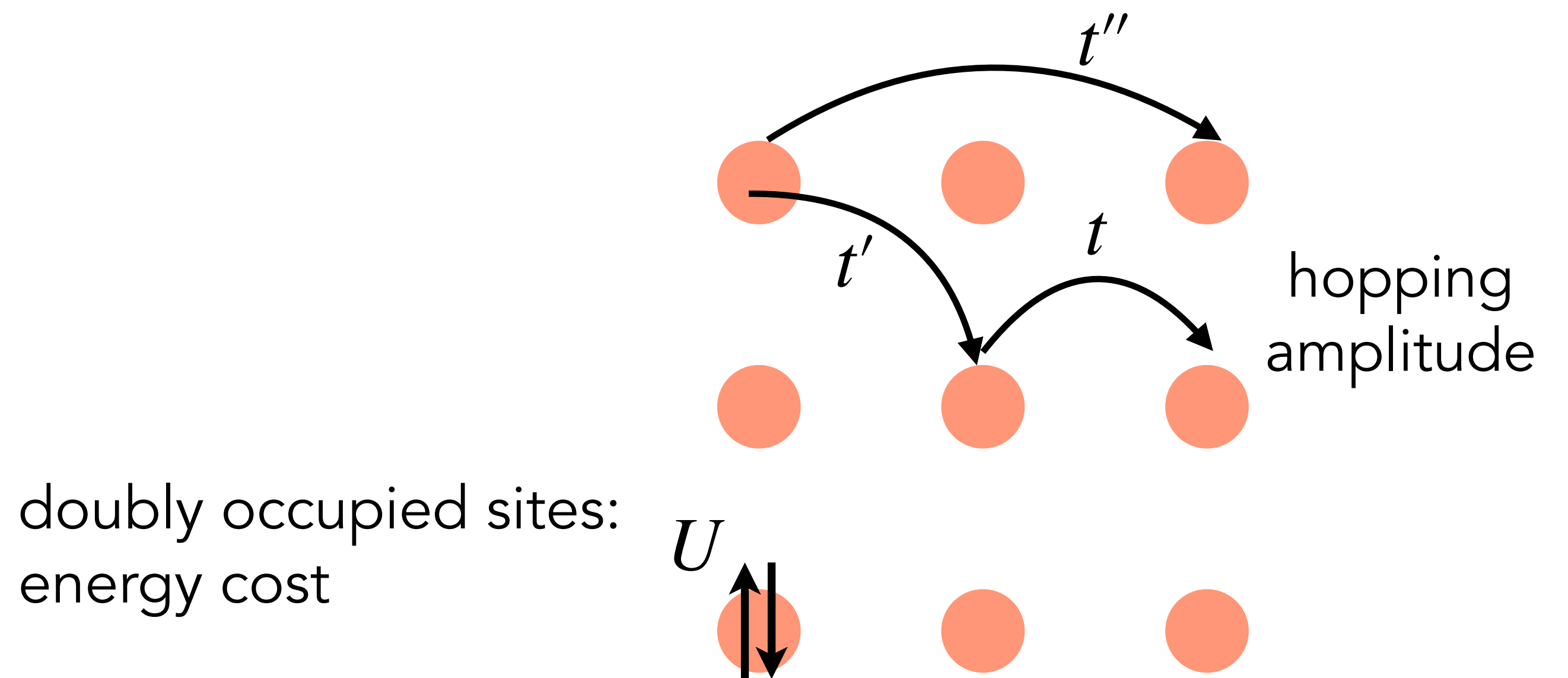
Understand **what the Hubbard model is** and **what we currently know** about it *before* trying to solve it with a quantum processor

Building a phase diagram

Parameters:

- Number of particles in the model
- Value of U
- Value of the hopping parameter(s)
- Lattice dimension (1D, 2D, ...)
- Number of lattice sites
- Temperature

$$H = \sum_{\langle ij \rangle, \sigma} t_{ij} c_{i\sigma}^\dagger c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}$$

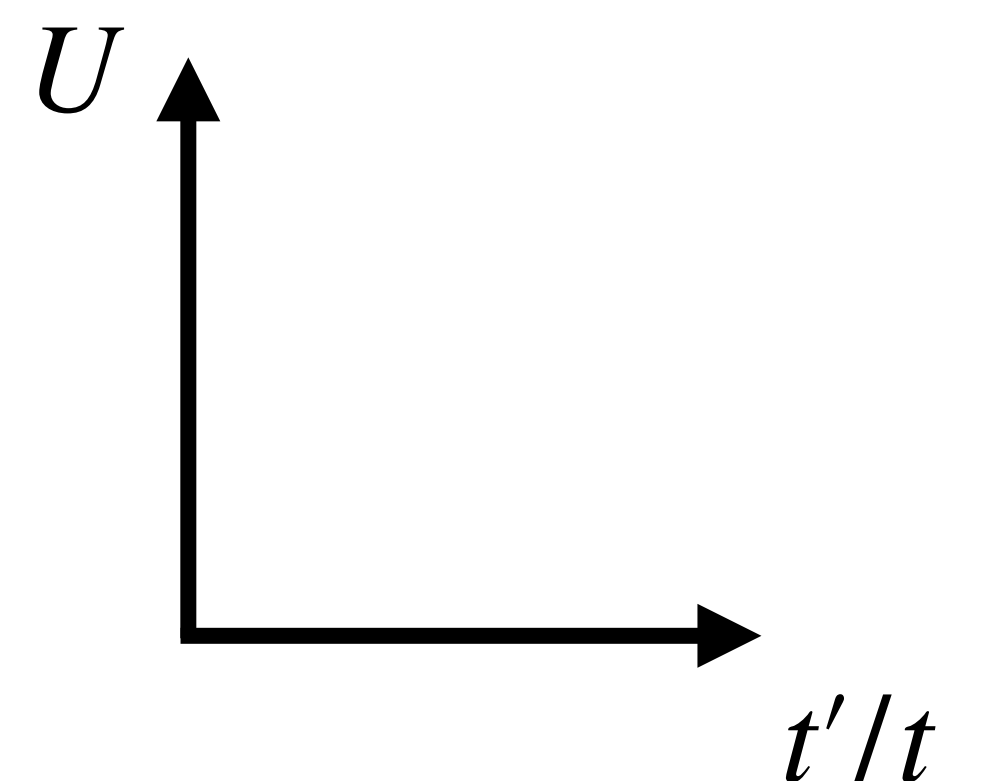
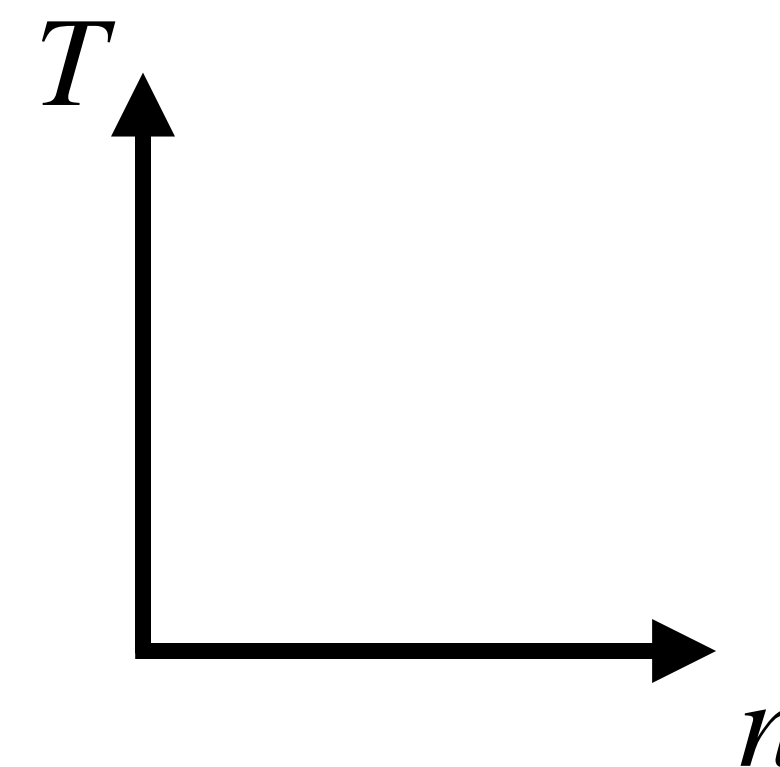
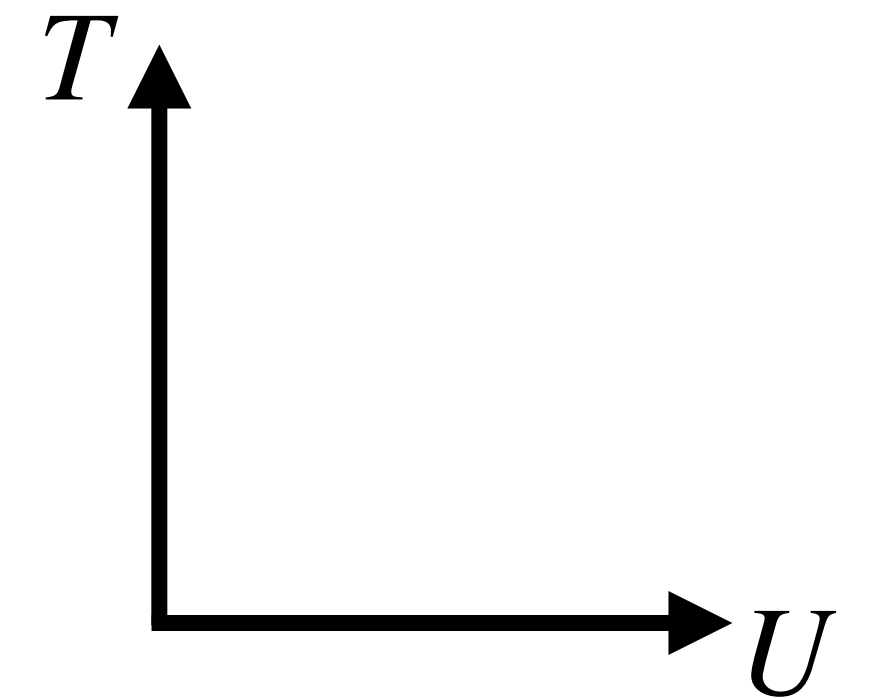
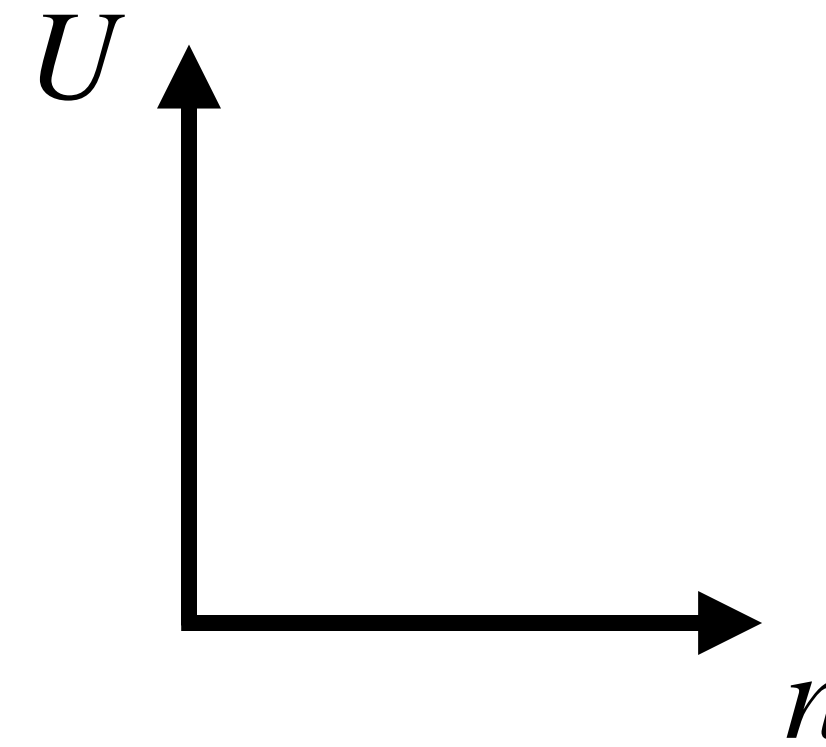


Building a phase diagram

Parameters:

- Number of particles in the model
- Value of U
- Value of the hopping parameter(s)
- Lattice dimension (1D, 2D, ...)
- Number of lattice sites
- Temperature

Phase diagrams



A zoo of classical approaches

Wavefunction methods:

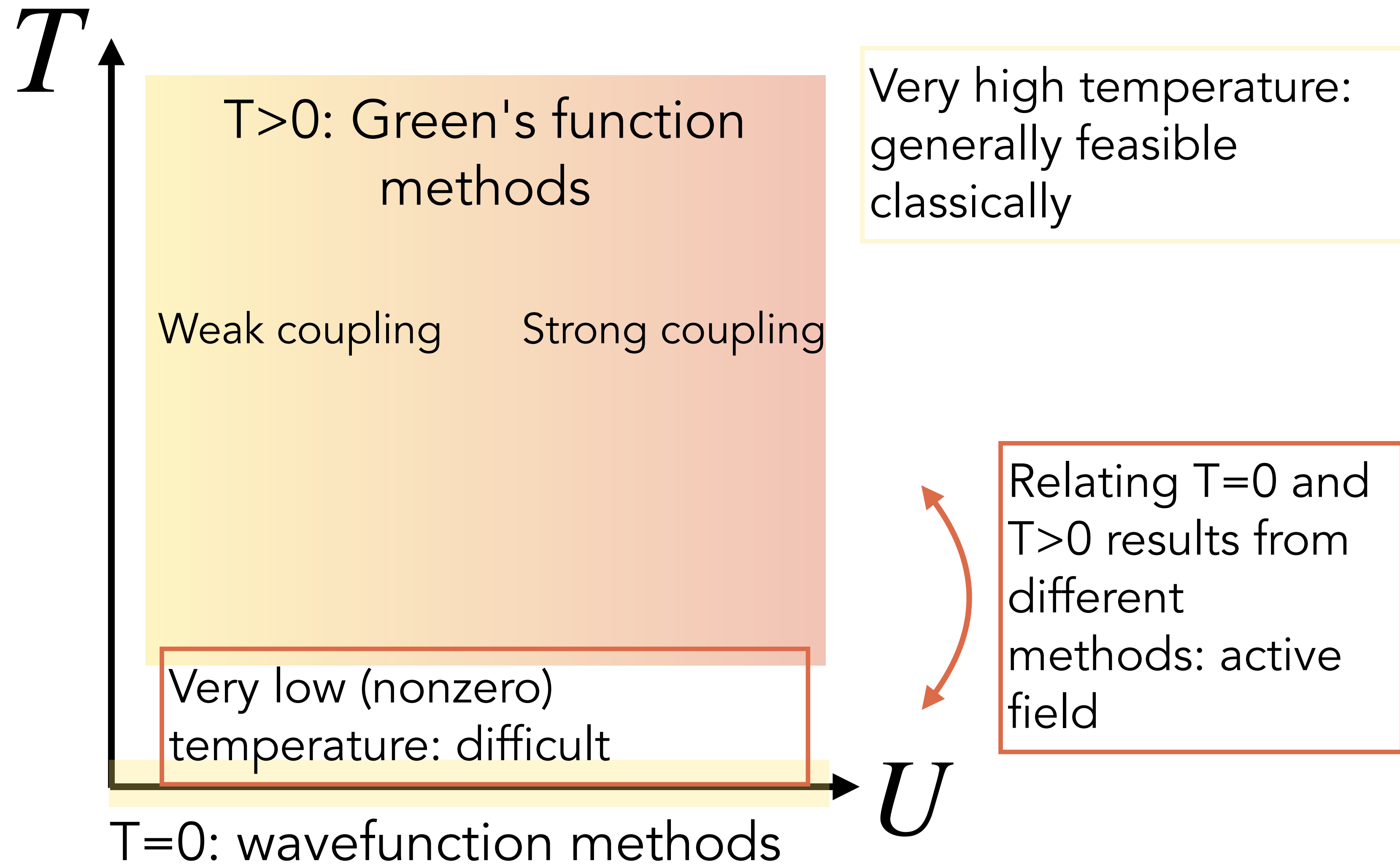
Exact diagonalization,
Variational Monte Carlo,
Tensor networks...

Green's function methods:

Dynamical mean-field theory,
Quantum Monte Carlo,
perturbation theory,

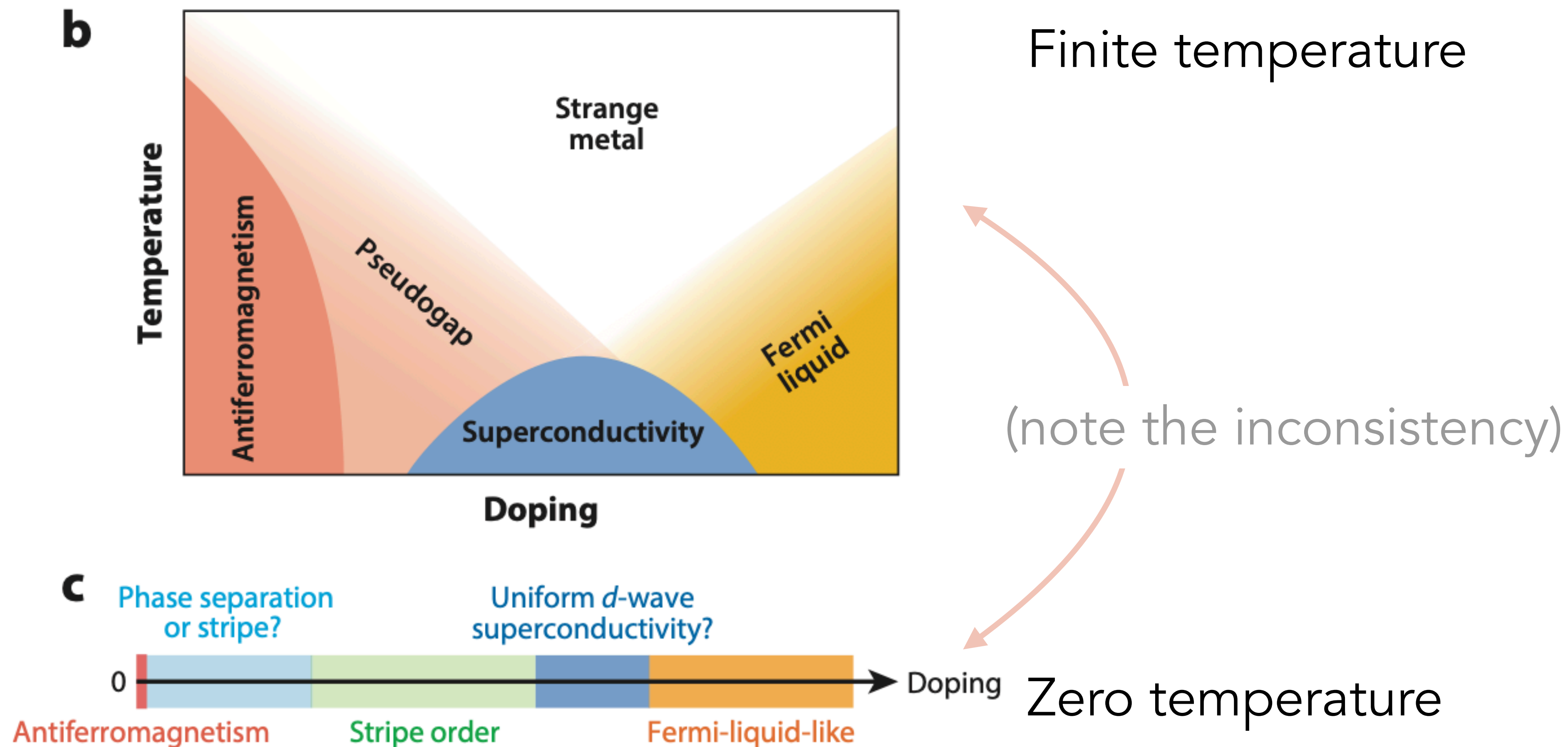
Reviews:

T. Schäfer *et al*, PRX **11**, 011058 (2021)
M. Qin *et al*, Annu.Rev.Condens.Matter Phys.
(2022)
D. Arovas *et al*, Annu.Rev.Condens.Matter
Phys (2022)



Some phase diagrams

2D Hubbard model



Example of a wavefunction approach: VMC

General idea

General problem: find the ground-state wavefunction of a given Hamiltonian

$$H|\Psi_{\text{GS}}\rangle = E_0|\Psi_{\text{GS}}\rangle$$

We choose an ansatz for the wavefunction

$$|\Psi_{\theta}\rangle$$

We optimize the θ parameters in order to find the lowest energy possible

$$E_{\theta} = \frac{\langle\Psi_{\theta}|H|\Psi_{\theta}\rangle}{\langle\Psi_{\theta}|\Psi_{\theta}\rangle} \geq E_0$$

The Hubbard model - VMC

Numerical
exercise

$$H = -t \sum_{\langle ij \rangle, \sigma} c_{i\sigma}^\dagger c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}$$

2 site Hubbard model

Write the hamiltonian in the following basis:

$$\rightarrow |\uparrow\downarrow, 0\rangle, |\uparrow, \downarrow\rangle, |\downarrow, \uparrow\rangle, |0, \uparrow\downarrow\rangle$$

Use the following ansatz for the wavefunction, with α a free variational parameter

$$\rightarrow |\psi_\alpha\rangle = e^{-\alpha} (|\uparrow\downarrow, 0\rangle + |0, \uparrow\downarrow\rangle) + |\uparrow, \downarrow\rangle + |\downarrow, \uparrow\rangle$$

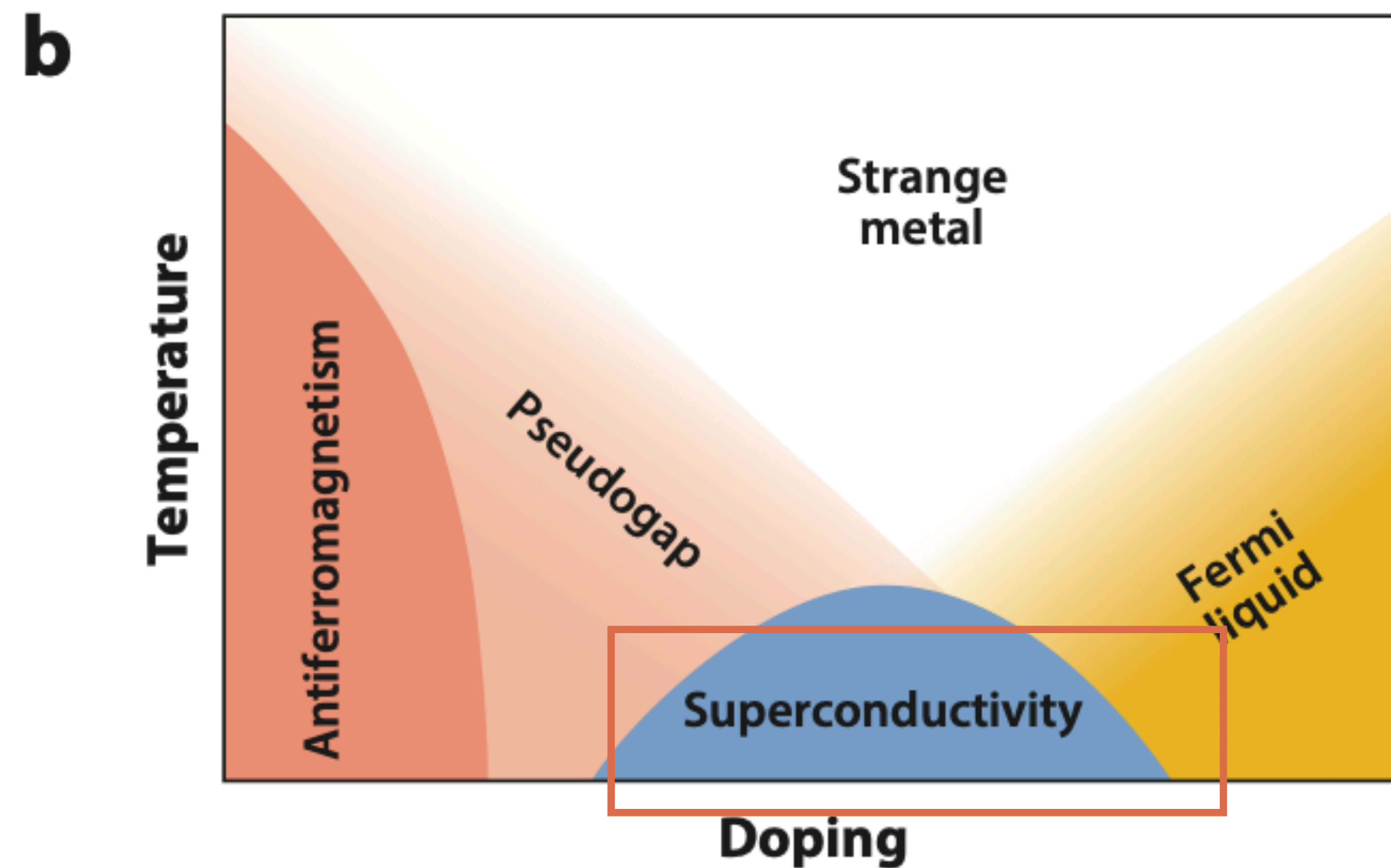
Minimize the ground state energy (scipy.optimize.minimize)

$$\rightarrow E_\alpha = \frac{\langle \psi_\alpha | H | \psi_\alpha \rangle}{\langle \psi_\alpha | \psi_\alpha \rangle}$$

4. Open questions

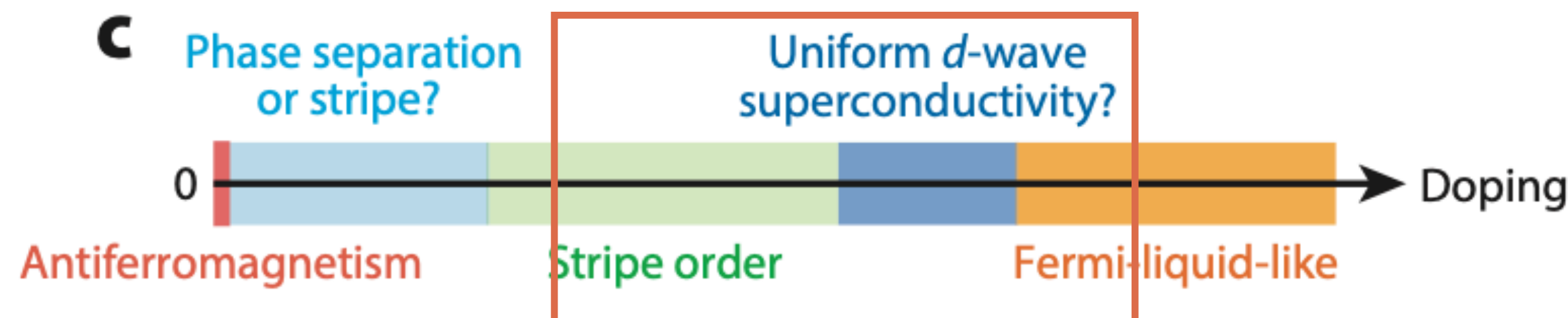
Open question 1: Superconductivity in 2D

2D Hubbard model



Finite temperature

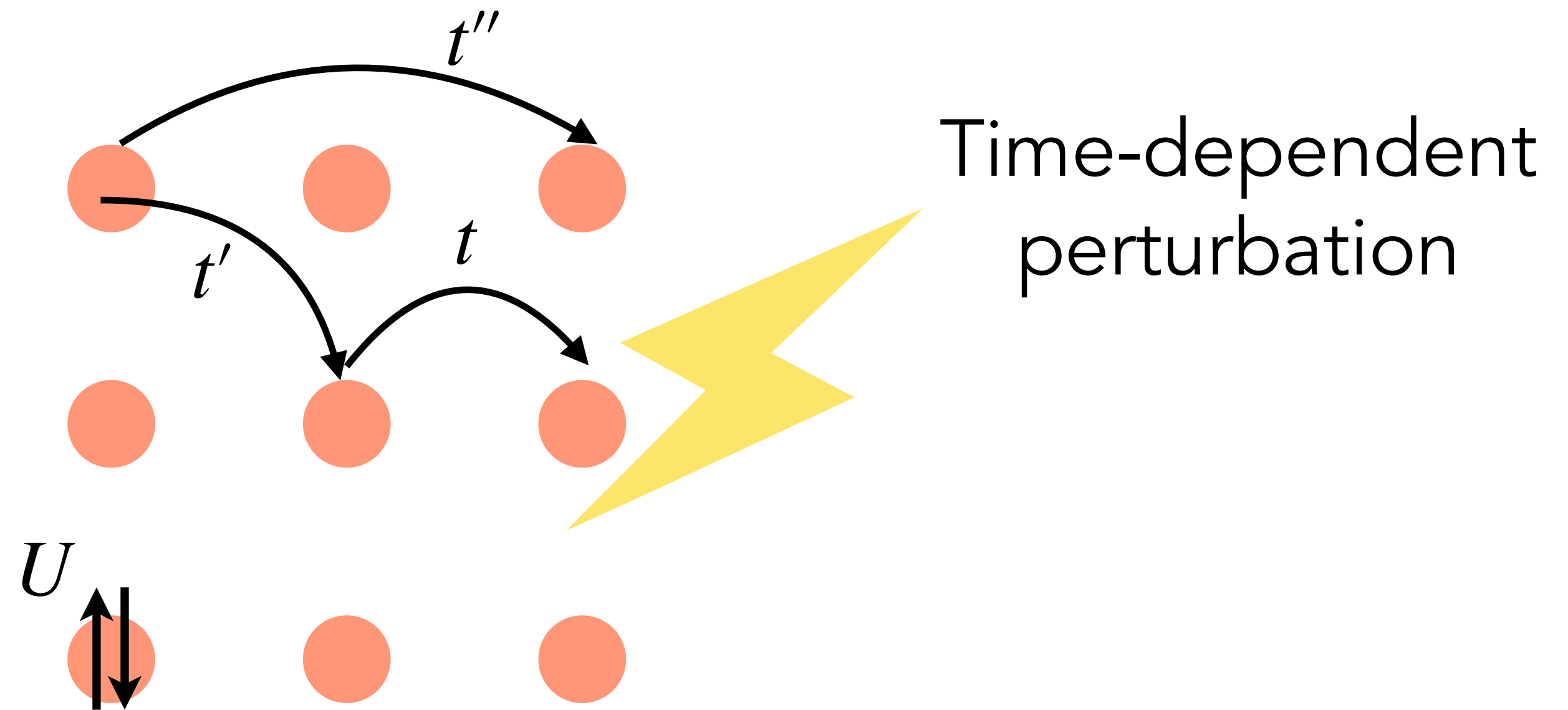
Whether or not there is d-wave SC in the ground state of the 2D Hubbard model is still debated.



Zero temperature

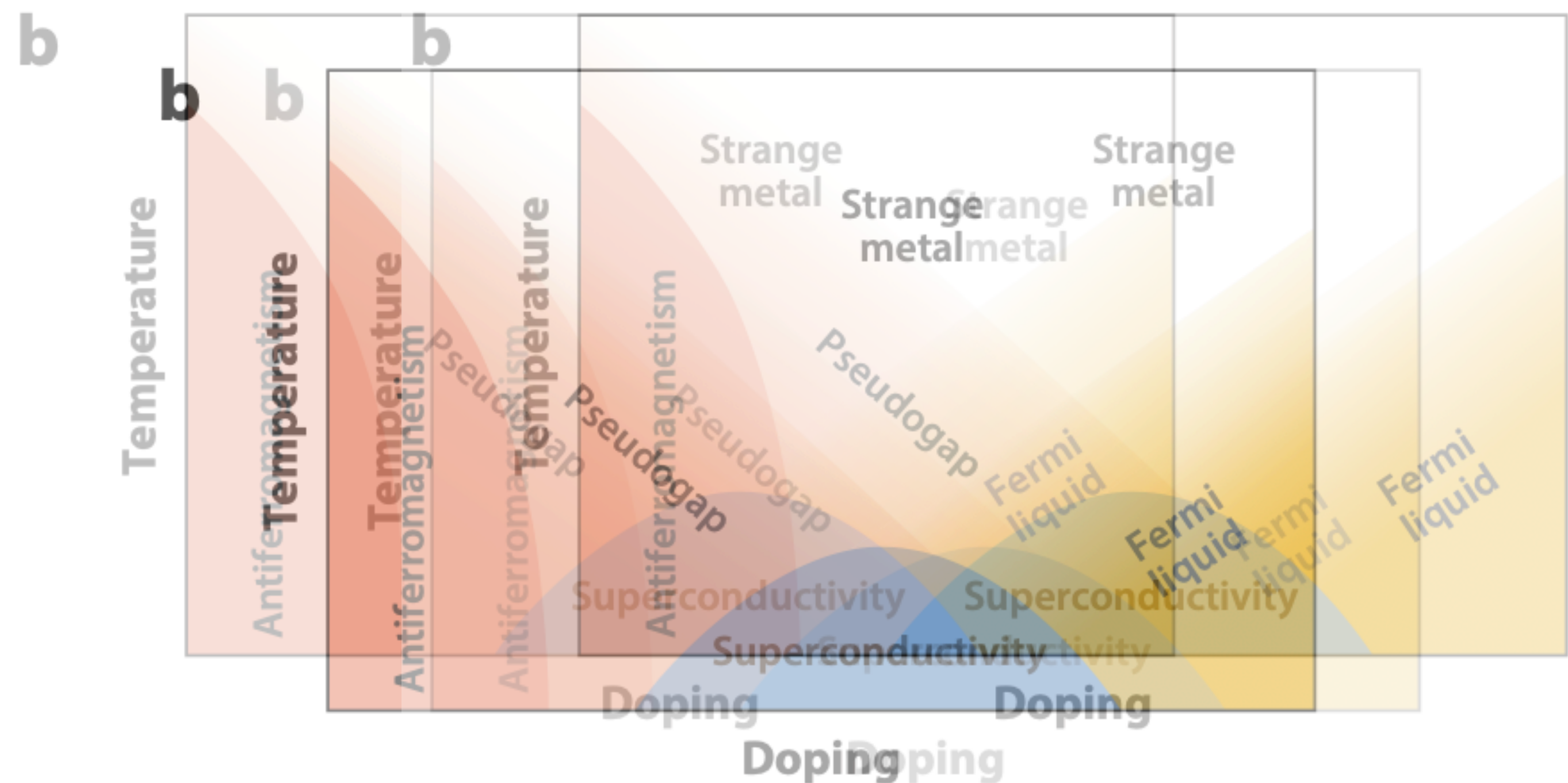
Open question 2: out of equilibrium

How does the system respond to perturbations that lead it **out of equilibrium**?



Example of an interesting question: generating new phases by driving the system out of equilibrium

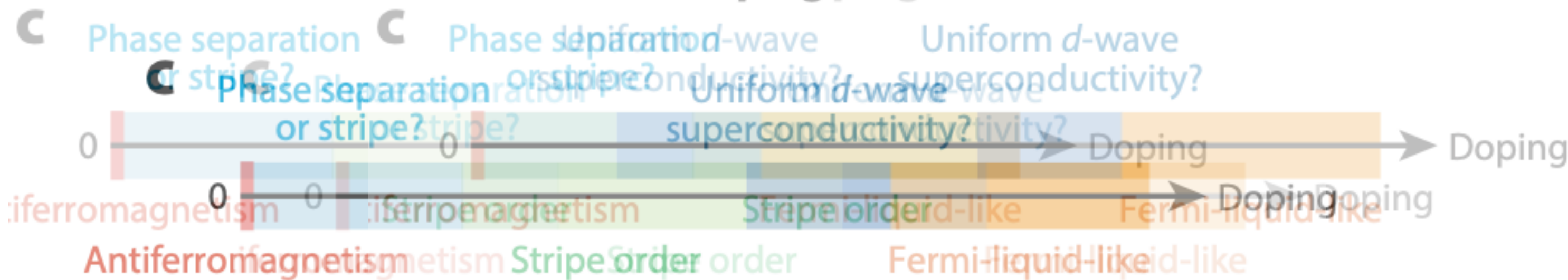
Open question 3: exact phase diagram boundaries



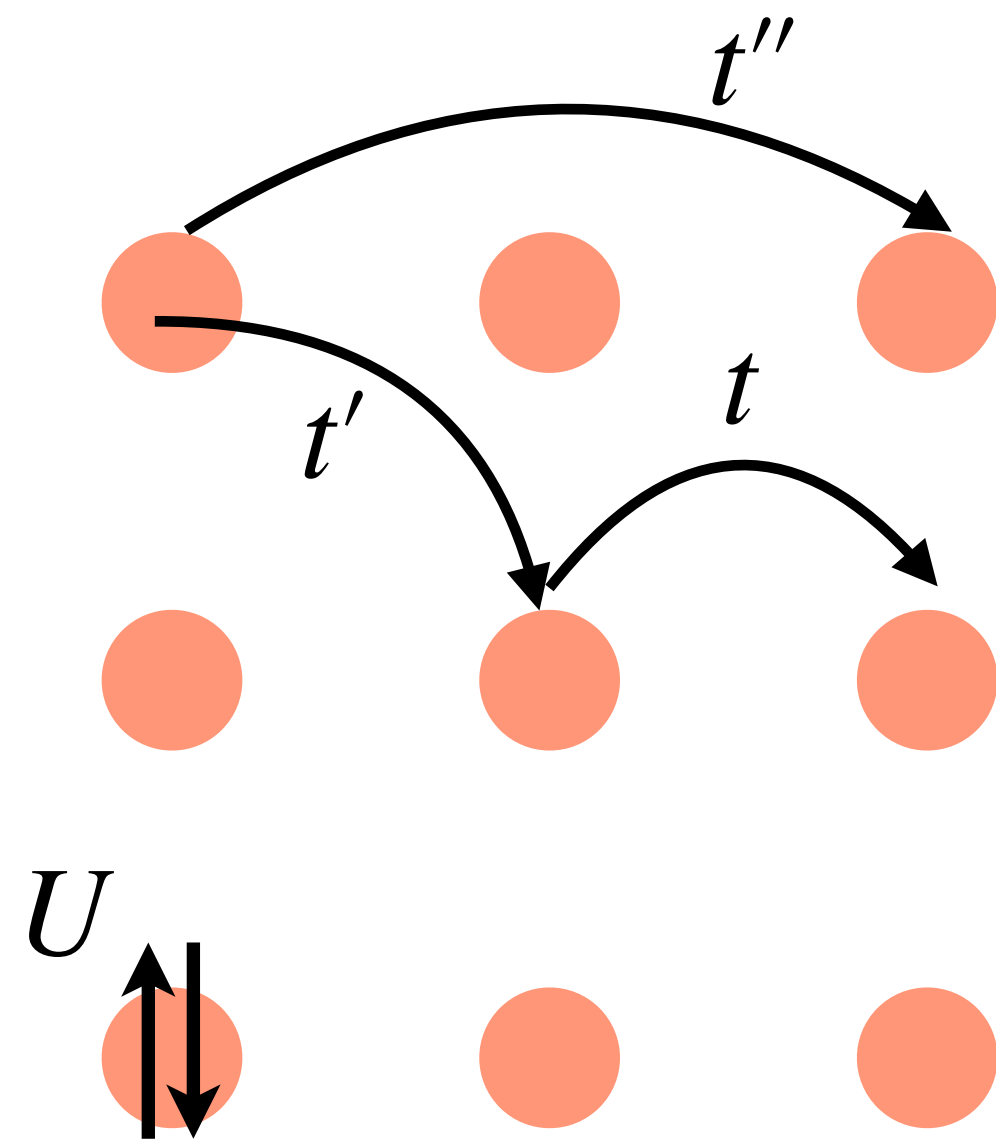
Superimposing the phase diagram obtained using different approaches: not always a perfect agreement

Need for reliable/
systematic benchmarks

T. Schäfer et al, PRX **11**, 011058 (2021)

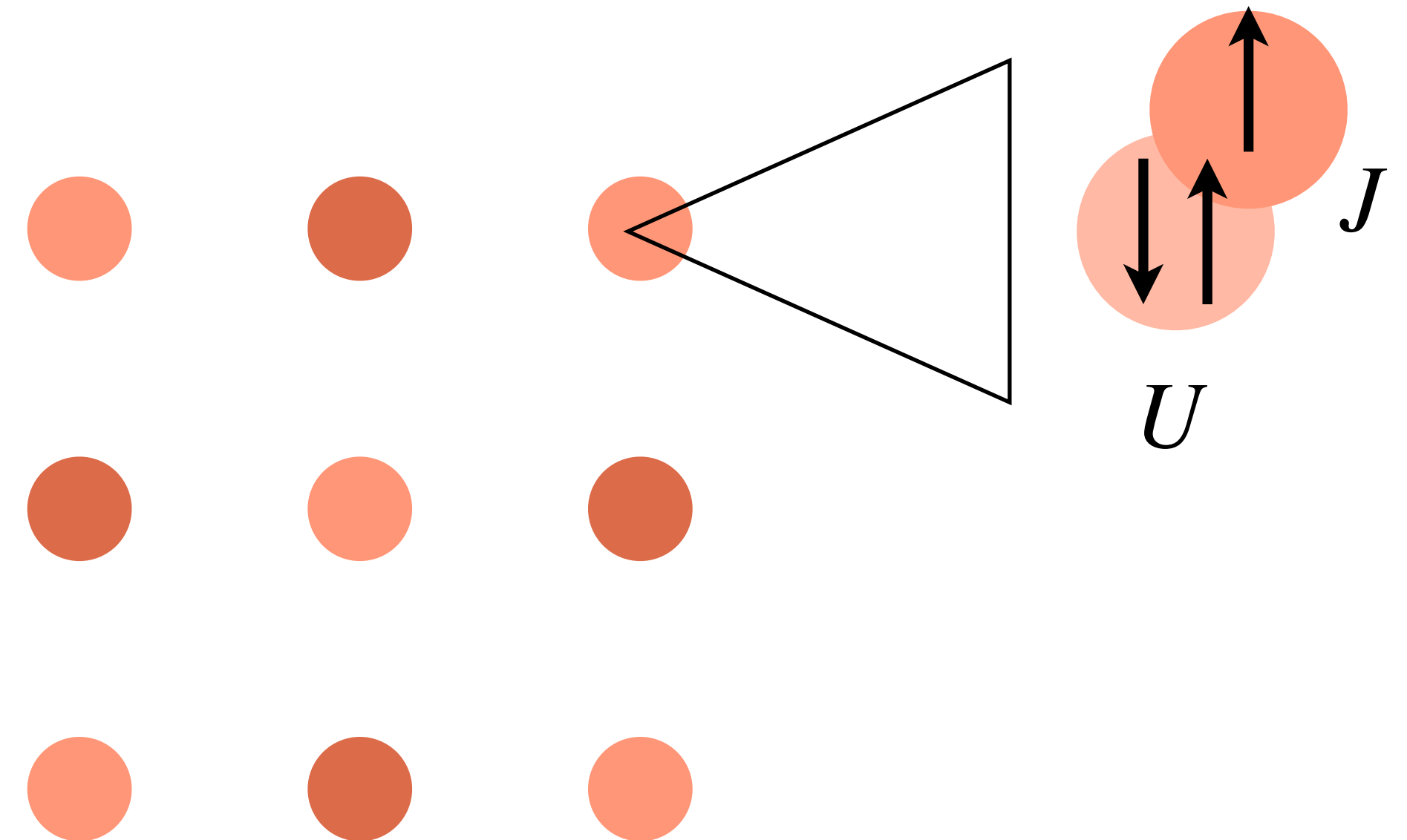


Open question 4: multiorbital systems



Hubbard model: one correlated orbital per site

$$c_{i\sigma}^\dagger, c_{i\sigma}$$



More general systems: multiple correlated and noncorrelated orbitals per site

$$c_{i\alpha\sigma}^\dagger, c_{i\alpha\sigma}$$

Introduction to the strongly-correlated electron problem

Lecture 2 - Quantum computing for the Hubbard model

Outline

Lecture 2

- Quantum processors seen as many-body systems
- From fermions to qubits
- Variational quantum eigensolver
- Impurity solvers

Goal of the lecture:

See how the Hubbard model can be **mapped** to a system of interacting qubits and explore two ways of **studying the model on a quantum processor**

Quantum processors as many-
body systems

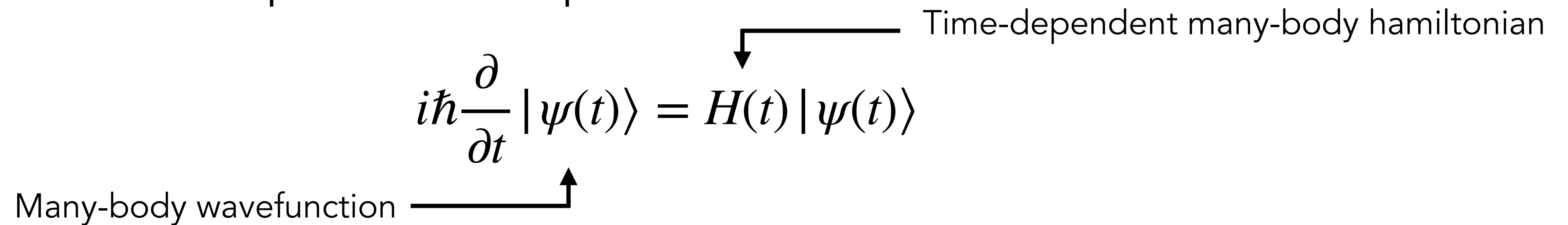
Many-body time evolution

Time evolution on a quantum computer

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = H(t) |\psi(t)\rangle$$

Time-dependent many-body hamiltonian

Many-body wavefunction



Where $|\psi(t=0)\rangle$ and $H(t)$ can be controlled/tuned for specific applications

Analog quantum computers

$H(t)$ is the hamiltonian of the system itself: Hubbard (bosons or fermions), Heisenberg, Ising...

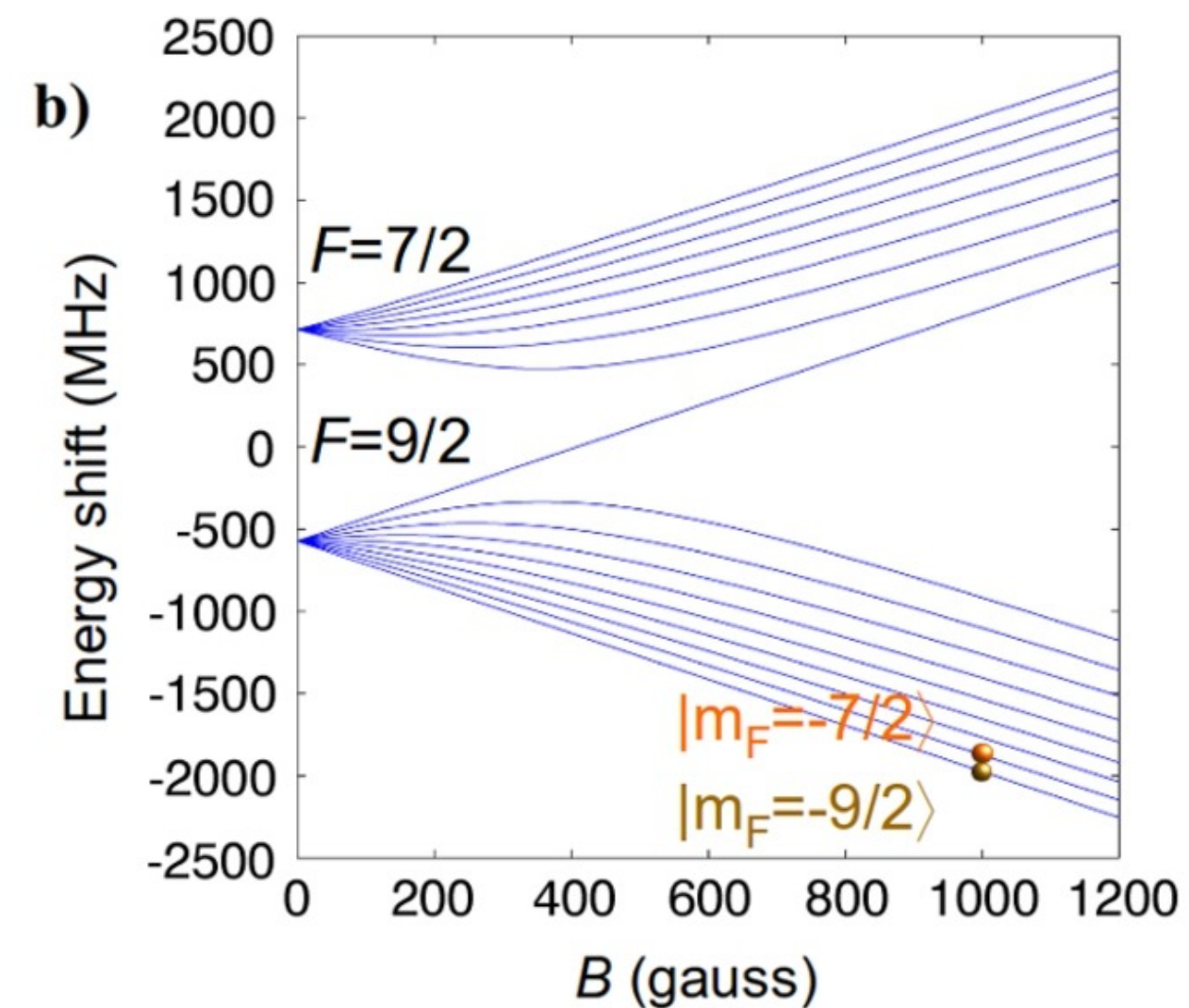
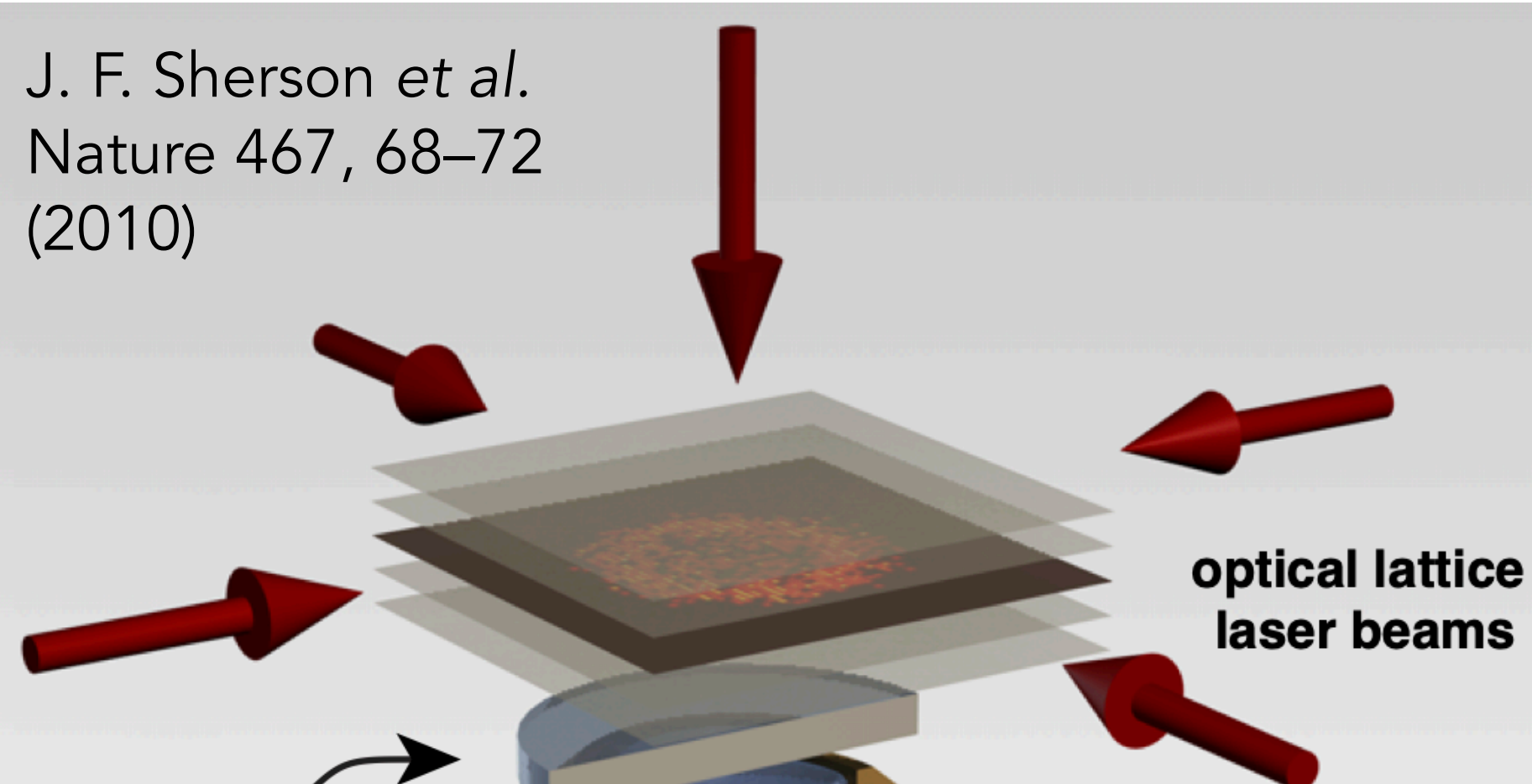
Gate-based quantum computers

Implement arbitrary unitary operations with gates

$$U = T e^{-i \int H(\tau) d\tau}$$

Ultracold atoms

J. F. Sherson *et al.*
Nature 467, 68–72
(2010)



Elements

A gas of fermionic isotopes

Laser beams forming an interference pattern

- Atoms are trapped in a lattice (variable geometry)

A magnetic field

- Lifts the degeneracy of the hyperfine states to get two states mapped to the fermionic spins

Resulting hamiltonian: Hubbard model

Model parameters can be tuned (example: t/U vs laser beam intensity)

Examples of phenomena observed with cold atoms

- The electron-electron interaction driven metal to insulator transition (Mott transition)

R. Jördens *et al.* Nature 455, 204–207 (2008)
U. Schneider *et al.* Science 322, 1520–1525 (2008)
- Antiferromagnetic order

D. Greif *et al.* Science, 340 (2013), pp. 1307-1310
- Bad metallic transport

Brown *et al.*, Science 363, 379–382 (2019)
- Low-temperature regime

M. Xu *et al.* Nature 642, 909–915 (2025)
- Charge and spin fluctuations at the onset of the pseudogap

T. Chalopin *et al.* PNAS 123 (4) e2525539123 (2025)

Towards the realization of a phase diagram?

From fermions to qubits

A very simple mapping

$$H = -t \sum_{\langle ij \rangle, \sigma} c_{i\sigma}^\dagger c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}$$

Basic operations

$$c^\dagger |0\rangle = |1\rangle$$

$$c |0\rangle = 0$$

$$c^\dagger |1\rangle = 0$$

$$c |1\rangle = |0\rangle$$

Simple mapping

$$\left. \begin{array}{l} c_\sigma^\dagger \rightarrow \sigma^- \\ c_\sigma \rightarrow \sigma^+ \end{array} \right\}$$

$$\sigma^- |0\rangle = |1\rangle$$

$$\sigma^+ |0\rangle = 0$$

$$\sigma^- |1\rangle = 0$$

$$\sigma^+ |1\rangle = |0\rangle$$

So: simply replace the creation/annihilation operators with Pauli matrices

...but the anticommutation relations are not there

Second quantization

Anticommutation relations

$$\{c_\alpha^\dagger, c_\beta\} = c_\alpha^\dagger c_\beta + c_\beta c_\alpha^\dagger = \delta_{\alpha\beta}$$

$$\{c_\alpha^\dagger, c_\beta^\dagger\} = 0$$

$$\{c_\alpha, c_\beta\} = 0$$

Pauli matrices

Commute for $\alpha \neq \beta$!

$$\alpha \neq \beta : [\sigma_\alpha^\pm, \sigma_\beta^\pm] = 0 \Rightarrow \{\sigma_\alpha^\pm, \sigma_\beta^\pm\} = 2\sigma_\alpha^\pm \sigma_\beta^\pm$$

This is a problem: anticommutation relations are **essential** to correctly represent the fermions

Anticommutation fixed: Jordan-Wigner mapping

Our goal: find a mapping that correctly encodes the fermionic anticommutation relations

Example for two sites

For two sites, try:

$$c_1^\dagger \rightarrow \sigma_1^- \otimes I_2$$

$$c_2^\dagger \rightarrow \sigma_1^z \otimes \sigma_2^-$$

$$\begin{aligned}\{c_1^\dagger, c_2^\dagger\} &= (\sigma_1^- \sigma_1^z + \sigma_1^z \sigma_1^-) \otimes \sigma_2^- \\ &= (\sigma_1^- - \sigma_1^-) \otimes \sigma_2^- \\ &= 0\end{aligned}$$

General Jordan-Wigner mapping

$$c_i^\dagger \rightarrow \sigma_i^- \otimes \sigma_{i-1}^z \otimes \dots \otimes \sigma_1^z$$

$$c_i \rightarrow \sigma_i^+ \otimes \sigma_{i-1}^z \otimes \dots \otimes \sigma_1^z$$

Ensures the fermionic anticommutation relations are satisfied

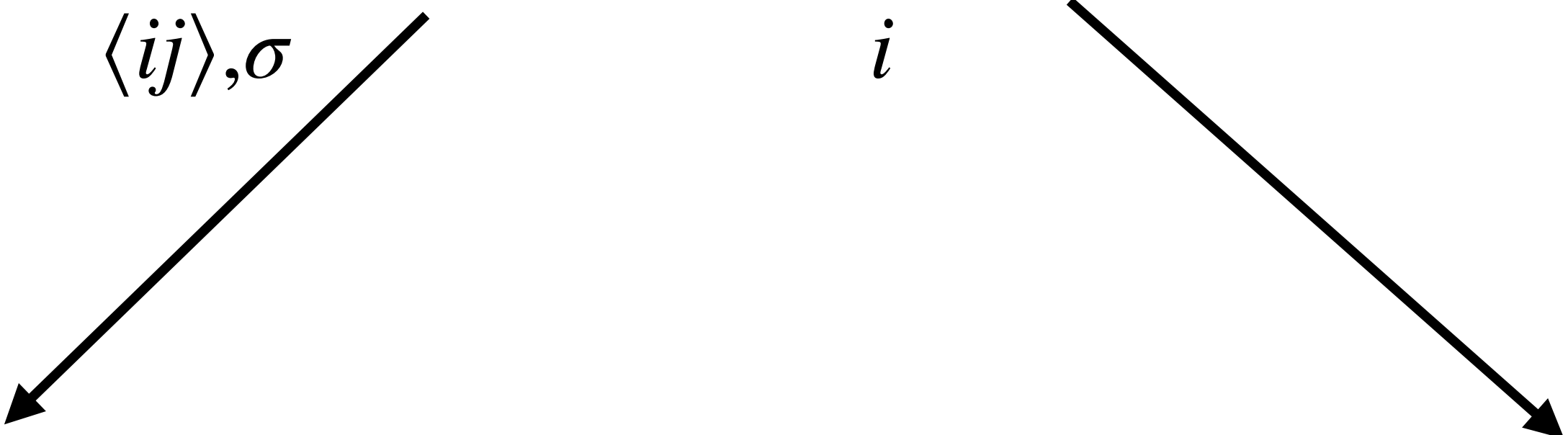
Requires **one qubit per fermionic mode** (site, spin, orbital...)

Rewriting the basis states

All single-site basis states	Mapping: two qubits
$ 0\rangle$	$ 00\rangle$
$ \uparrow\rangle$	$ 10\rangle$
$ \downarrow\rangle$	$ 01\rangle$
$ \uparrow\downarrow\rangle$	$ 11\rangle$

Half-filled two- particle states	Mapping: four qubits
$ \uparrow, \downarrow\rangle$	$ 1001\rangle$
$ \downarrow, \uparrow\rangle$	$ 0110\rangle$
$ \uparrow\downarrow, 0\rangle$	$ 1100\rangle$
$ 0, \uparrow\downarrow\rangle$	$ 0011\rangle$

Rewriting the hamiltonian

$$H = -t \sum_{\langle ij \rangle, \sigma} c_{i\sigma}^\dagger c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}$$


$$c_\alpha^\dagger c_\beta = \left(\sigma_\alpha^- \otimes \sigma_{\alpha-1}^z \otimes \dots \otimes \sigma_1^z \right) \left(\sigma_\beta^- \otimes \sigma_{\beta-1}^z \otimes \dots \otimes \sigma_1^z \right)$$

$$c_\alpha^\dagger c_\alpha = \frac{1}{2} (I - \sigma_\alpha^z)$$

The hamiltonian becomes a sum of Pauli strings with different coefficients

$$\Rightarrow H = \sum_\lambda b_\lambda h_\lambda$$

Variational quantum eigensolver

Reminder: variational principle

General idea

General problem: find the ground-state wavefunction of a given Hamiltonian

$$H|\Psi_{\text{GS}}\rangle = E_0|\Psi_{\text{GS}}\rangle$$

We choose an ansatz for the wavefunction

$$|\Psi_{\theta}\rangle$$

We optimize the θ parameters in order to find the lowest energy possible

$$E_{\theta} = \frac{\langle\Psi_{\theta}|H|\Psi_{\theta}\rangle}{\langle\Psi_{\theta}|\Psi_{\theta}\rangle} \geq E_0$$

Variational quantum eigensolver

General overview

Wave function ansatz

$$|\psi_\theta\rangle = U_\theta |0\rangle$$

Hamiltonian in terms of Pauli operators

$$H = \sum_{\lambda} b_{\lambda} h_{\lambda}$$

Single Pauli string or multiple commuting Pauli strings

Measurement

$$\langle\psi_\theta| \dots X_i Y_j Z_k \dots |\psi_\theta\rangle$$

Measure Pauli Z on qubit k , etc.
N shots

Classical optimizer

$$E_\theta = \sum_{\lambda} \langle\psi_\theta| h_{\lambda} |\psi_\theta\rangle \Rightarrow \theta' \Rightarrow U(\theta')$$

Update the variational parameters

Exercise: 2-site Hubbard model

- Take the following mapping between the four qubits and the four fermionic modes:

$$q_0 = 1 \uparrow, q_1 = 2 \uparrow, q_2 = 1 \downarrow, q_3 = 2 \downarrow$$

- Write the Hubbard hamiltonian with the Jordan-Wigner transformation
- Discuss whether the following ansatz is an appropriate choice for a VQE simulation of this problem:

$$|\psi_\theta\rangle = \cos(\theta) |1001\rangle + \sin(\theta) |0110\rangle$$

Half-filled two-particle states	Mapping: four qubits
$ \uparrow, \downarrow\rangle$	$ 1001\rangle$
$ \downarrow, \uparrow\rangle$	$ 0110\rangle$
$ \uparrow\downarrow, 0\rangle$	$ 1010\rangle$
$ 0, \uparrow\downarrow\rangle$	$ 0101\rangle$

Practical considerations

- Balancing the expressiveness of the ansatz and the optimization process
Common issue with variational approaches: getting stuck in parameter regions away from the true ground state
- Finding the proper ansatz
- Overhead due to the number of shots required
- Limited to small systems on today's devices

Beyond the ground state: time
evolution

Reminder: strong correlations vs DFT

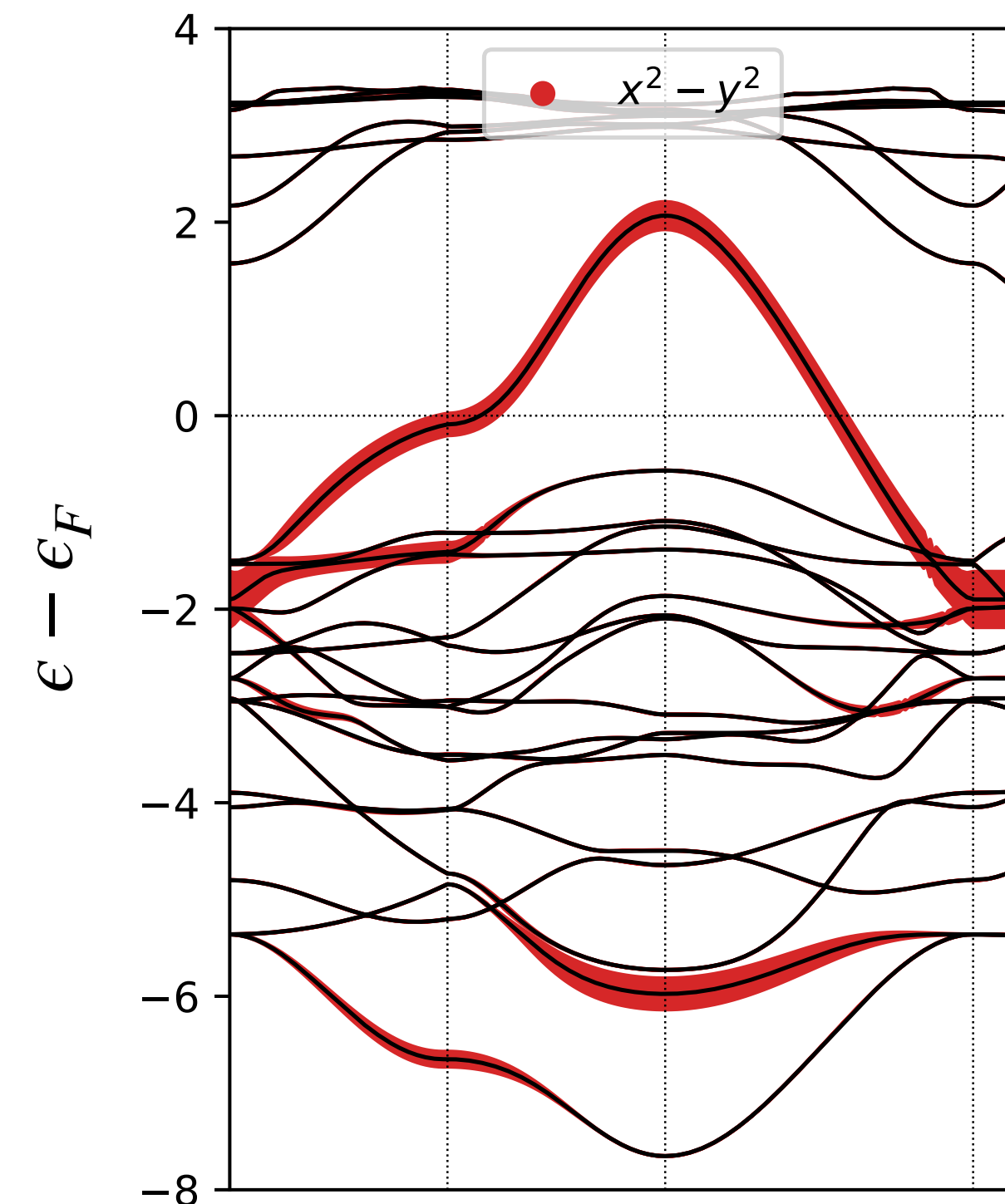
What is missing

Missing here: a proper treatment of electron-electron interactions

Definition: strong correlations

Strong electronic correlations: effects from electron-electron interactions that are not treated at the DFT level.

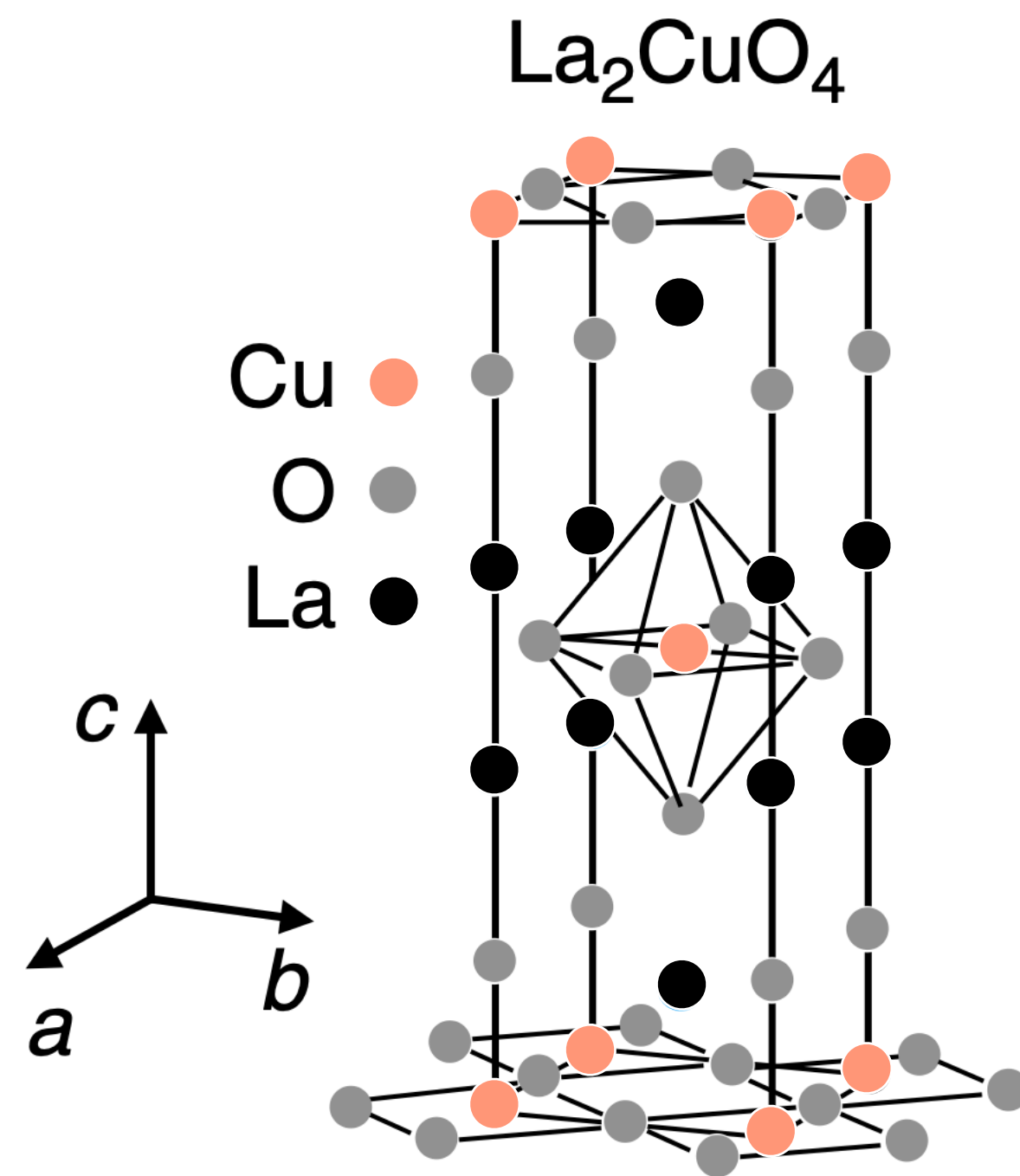
DFT output: ground state electronic properties



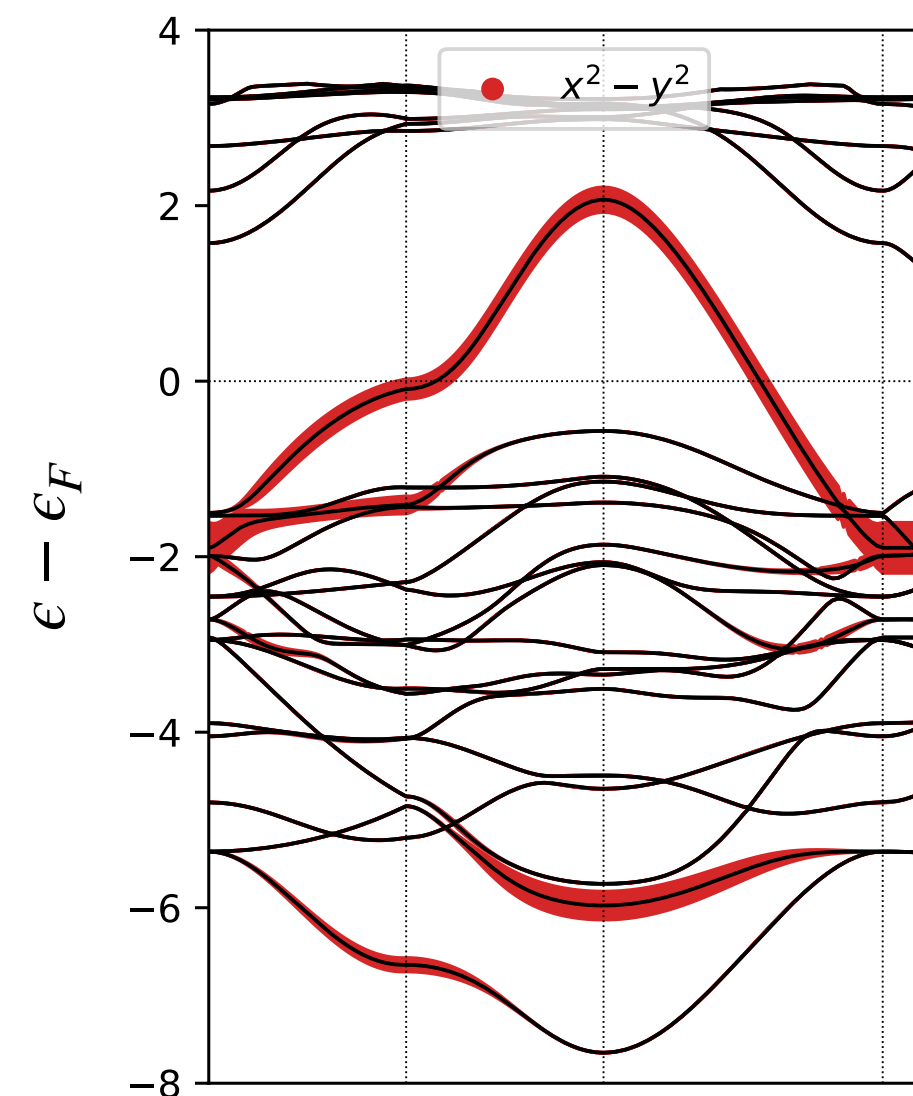
Metallic ground state

A classical framework to treat correlations better

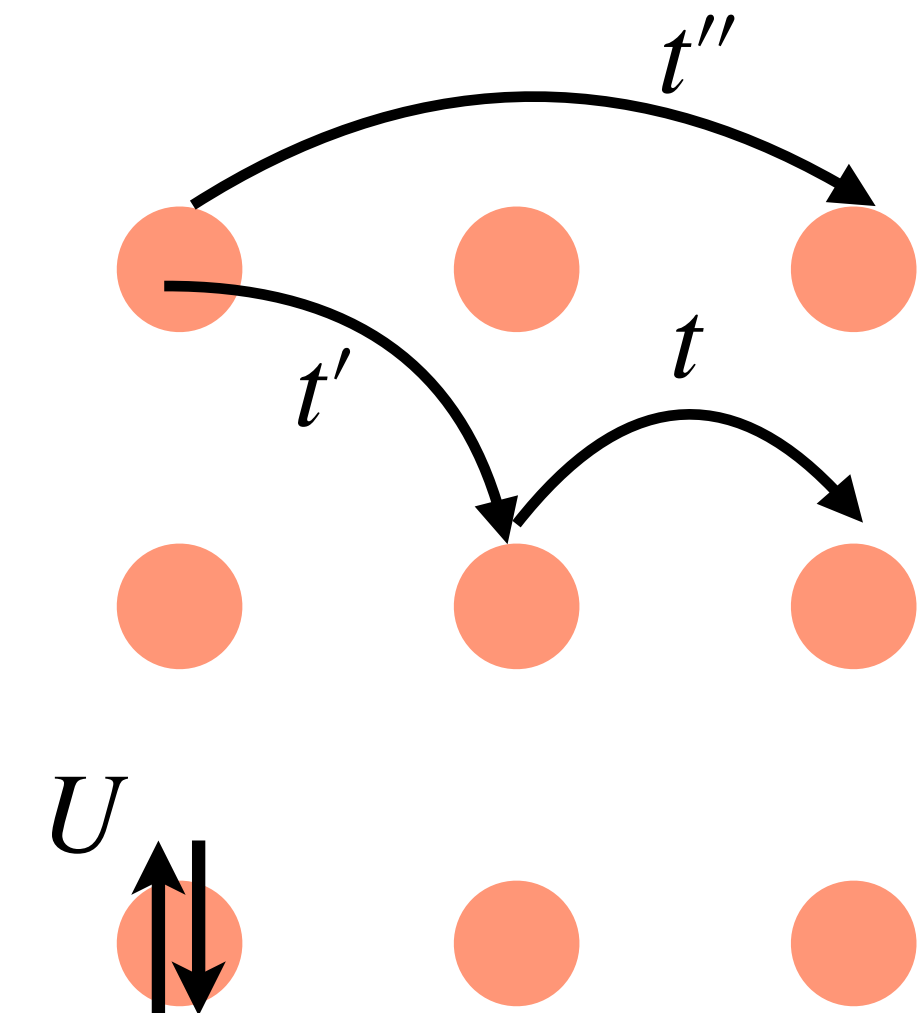
Input: crystal structure



DFT: ground state properties, incomplete treatment of correlations



Better treatment of the correlated subspace: Hubbard model simulation

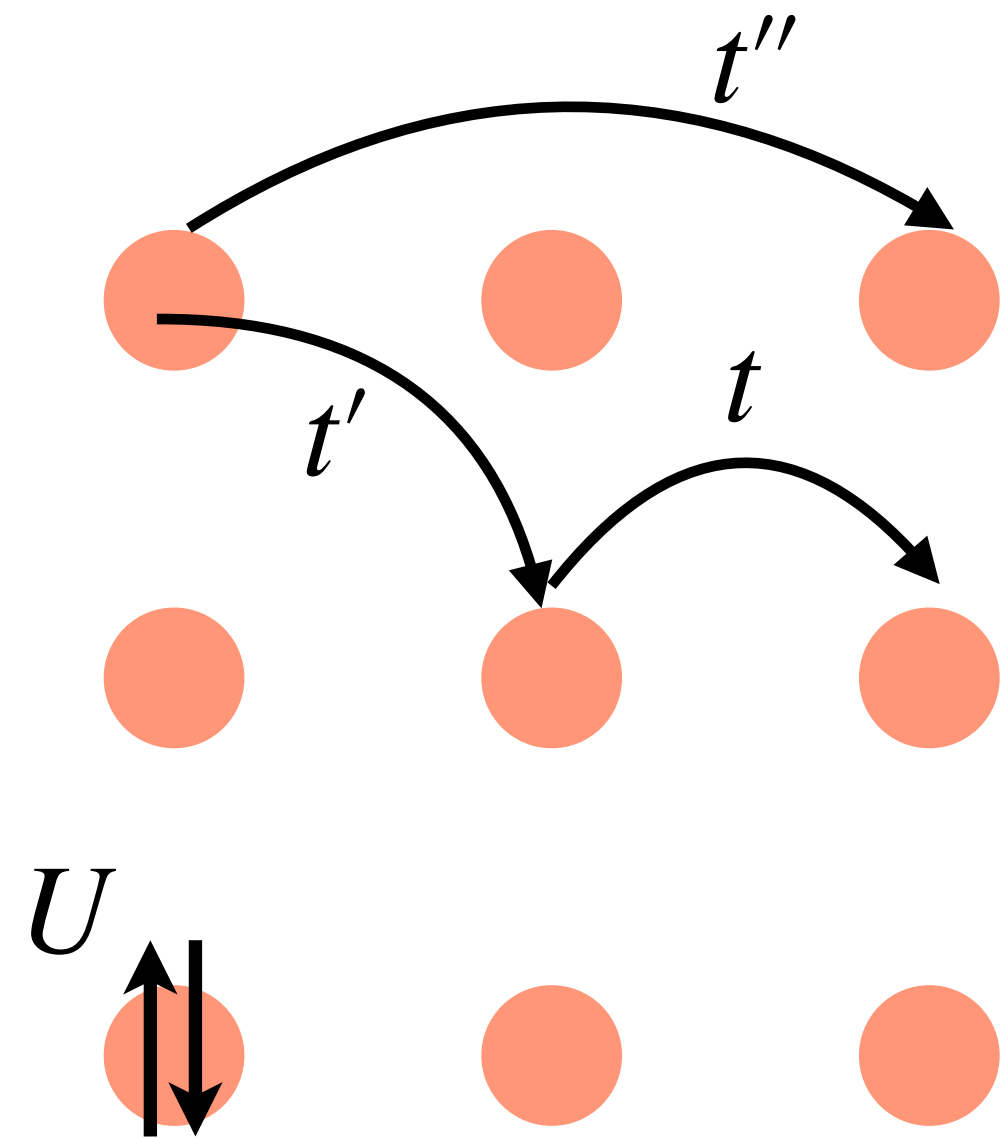


M.-E. Boulanger *et al.*
Nature Communications 11, 5325 (2020)

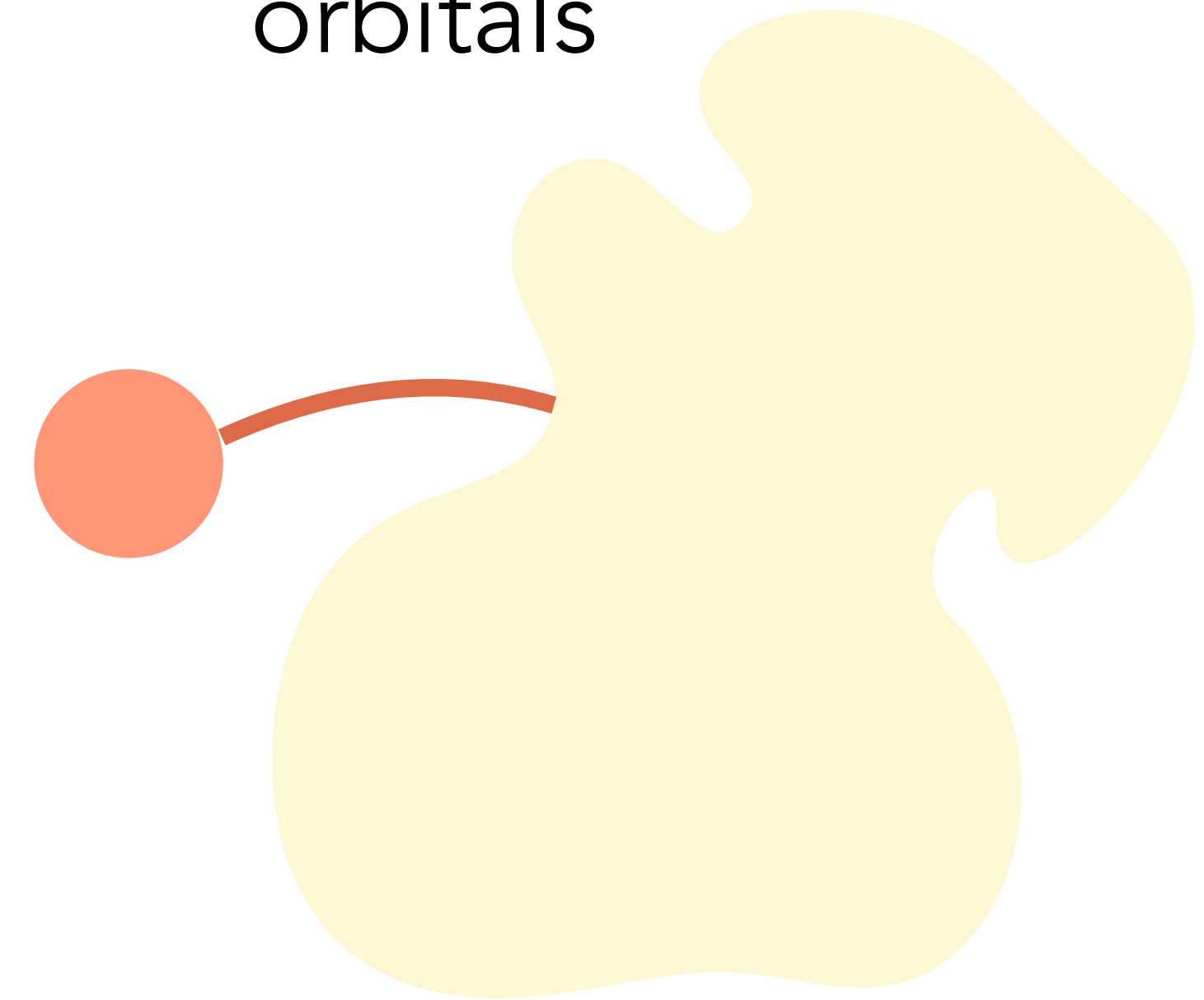
⇒ Realistic simulation of strongly correlated systems

Mapping to an impurity problem: DMFT

Lattice of correlated orbitals



One correlated orbital coupled
to a bath of noninteracting
orbitals

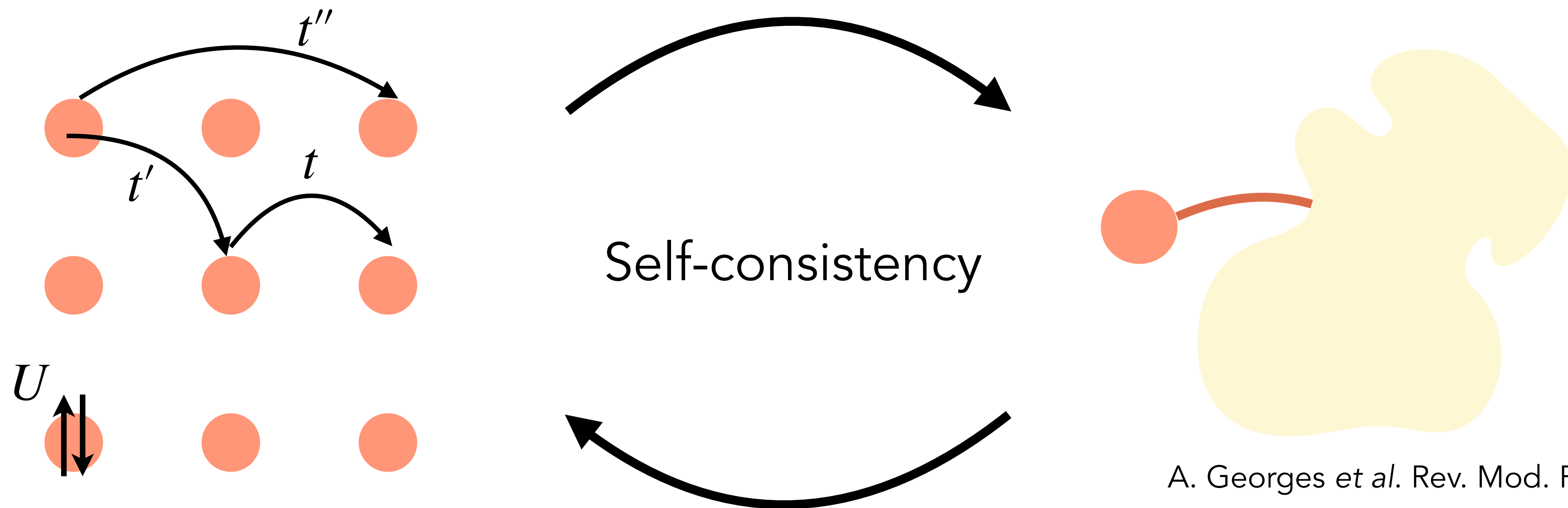


Self-consistency

We solve this problem and input its
solution back into the full lattice

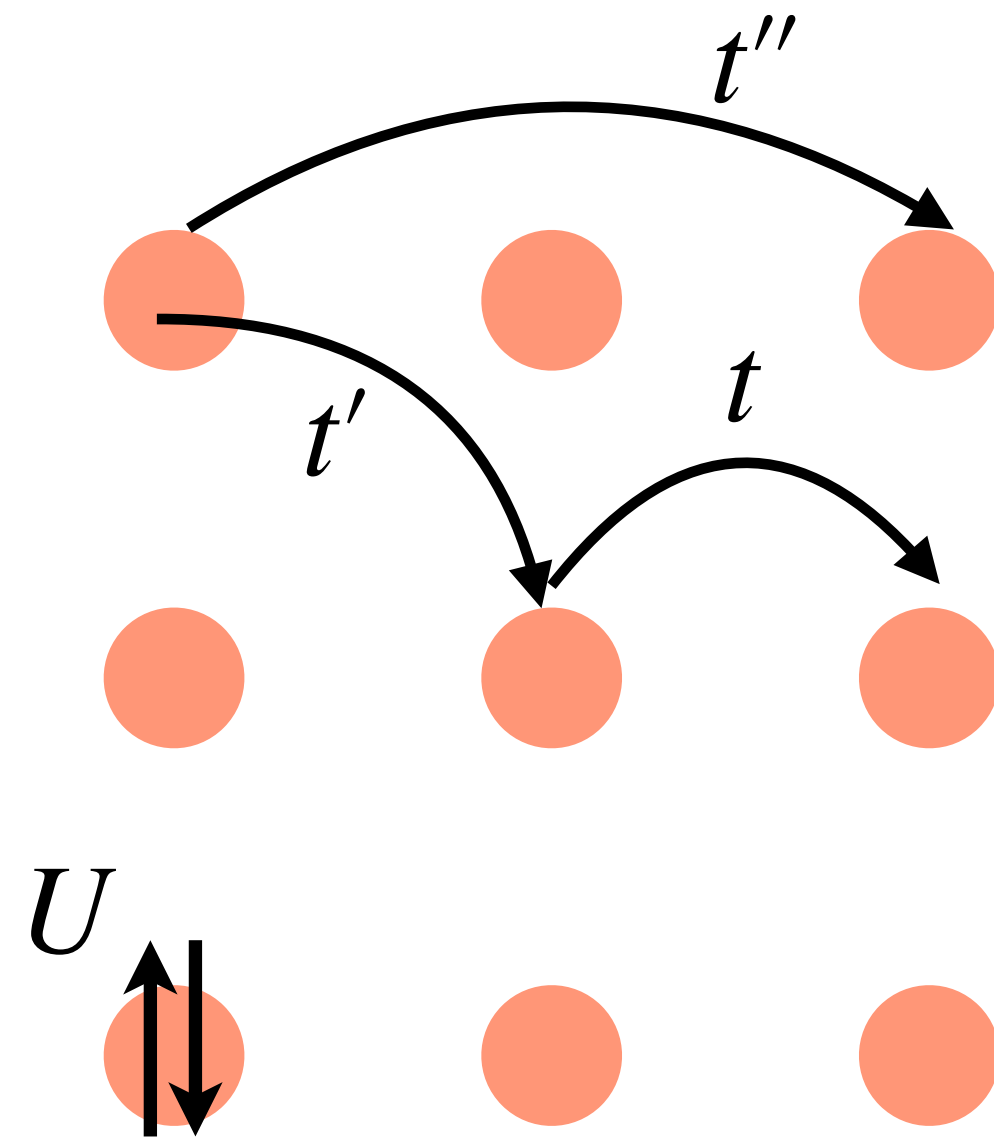
Successes of DMFT

- Exact solution of the Hubbard model in infinite dimension
- Description of the metal to insulator transition due to electronic correlations (Mott transition)
- Paired with DFT: realistic description of strongly correlated systems

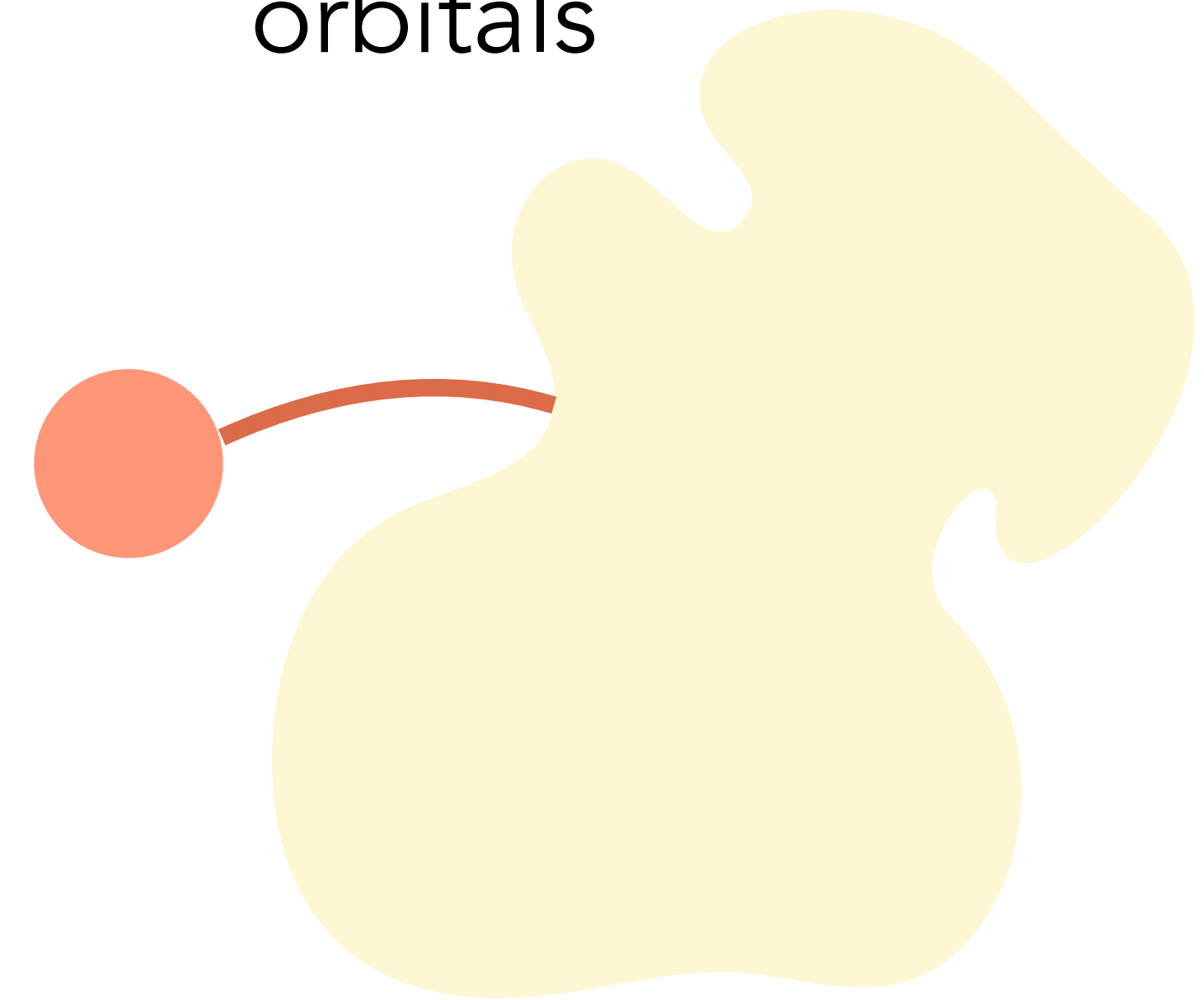


Mapping to an impurity problem: DMFT

Lattice of correlated orbitals



One correlated orbital coupled
to a bath of noninteracting
orbitals



Self-consistency

$$G_{\text{latt}} = G_{\text{imp}}$$

Green's functions

Solving the impurity problem in real time is still a difficult task.

Goal: use a quantum computer to solve the impurity problem $\Rightarrow$ obtain G_{imp}

Green's functions (math)

Differential equation:

$$\hat{L}u(x) = f(x)$$

↑ Linear differential operator (example: d/dx)

Definition of the Green's function:

$$\hat{L}G(x, s) = \delta(x - s)$$

Motivation:

$$\int ds \hat{L}G(x, s)f(s) = \int ds \delta(x - s)f(s) = f(x)$$

$$= \hat{L} \int ds G(x, s)f(s) = f(x)$$

$$= \hat{L} \int ds G(x, s)f(s) = \hat{L}u(x)$$

$$\Rightarrow u(x) = \int ds G(x, s)f(s)$$

Green's functions help us solve differential equations.

Green's functions (many-body problem)

Differential equation:

$$i\partial_t\psi(x, t) = H\psi(x, t)$$

Definition of the Green's function (**no interactions**):

True, mathematical Green's function

$$\hat{L}G_0(x, s; t, t') = -\delta(x - s)\delta(t - t')$$

Time-dependent Schrödinger equation: $i\partial_t - H_0$

Definition of the Green's function (**with interactions**):

$$G^{-1}(x, t) = G_0^{-1}(x, t) - \Sigma(x, t)$$

All the effects of electron-electron interactions

Green's functions help us understand the evolution of interacting particles.

Solving the impurity problem

Green's function to compute

$$G^>(t) = -i \langle \Psi_0 | e^{iH_{\text{imp}}t} c_\sigma e^{-iH_{\text{imp}}t} c_\sigma^\dagger | \Psi_0 \rangle$$

Time evolution

Ground state wavefunction

Gate-based quantum computer:

- Jordan-Wigner mapping
- Trotter decomposition of the time-evolution operators

Circuits:

$$\langle \Psi_0 | V^\dagger P_i V P_j | \Psi_0 \rangle$$

Pauli strings

Trotterized time-evolution

Conclusion

Useful resource for many-body simulations: the **TRIQS** library.

triqs.github.io

Lecture 1

- Strongly-correlated electrons are still systems of interest
- Many classical approaches exist, and many are efficient (or even numerically exact!)
- But there are still open questions

Lecture 2:

- The Hubbard model can be mapped to systems of qubits
- Different ways in which a quantum computer could be used: state preparation, time evolution, etc.

Efficient/useful implementations of algorithms for the Hubbard model on quantum computers will require collaboration between researchers of both fields.